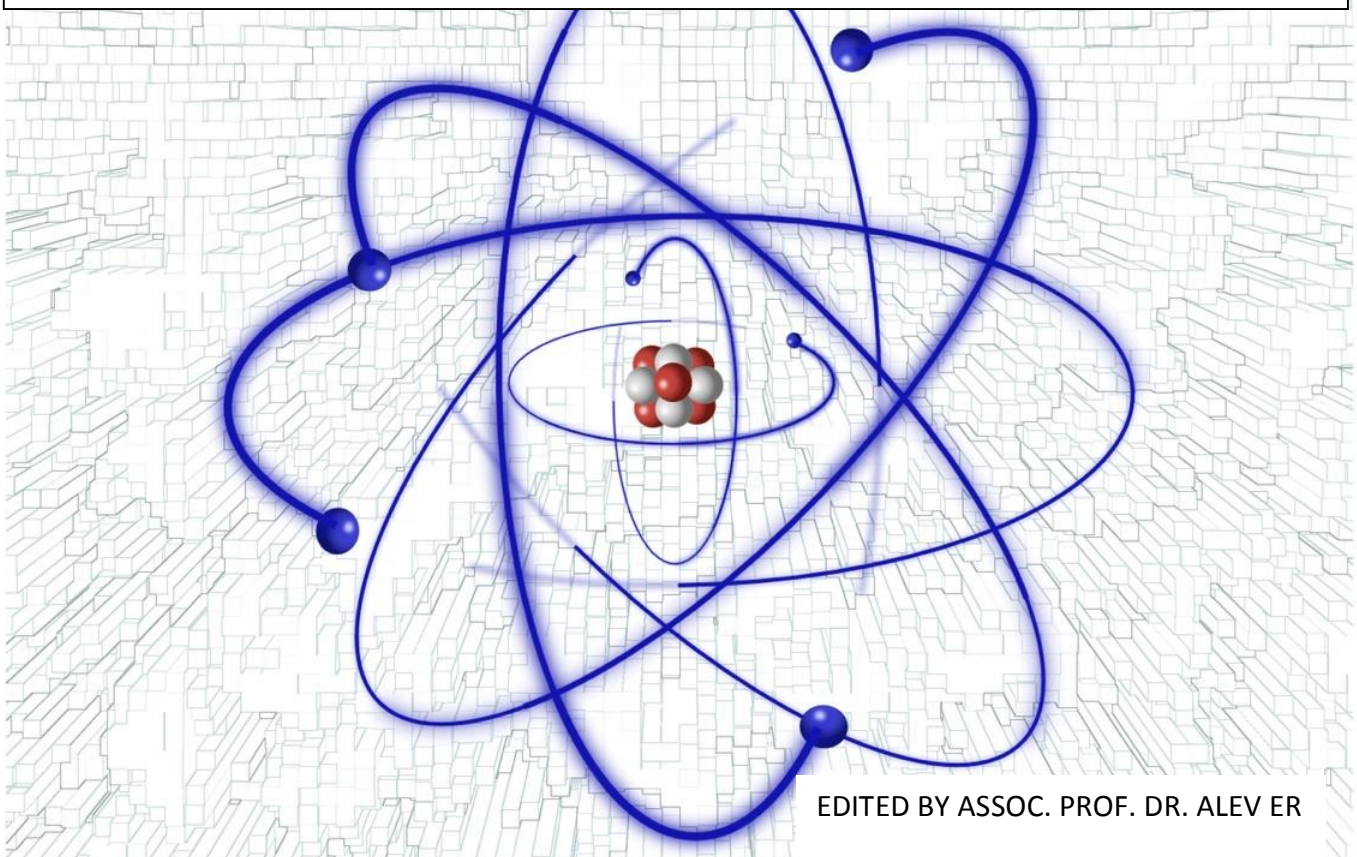
T.C.
İSTANBUL ÜNİVERSİTESİ
FEN FAKÜLTESİ
FİZİK BÖLÜMÜ

ATOMIC AND MOLECULAR PHYSICS LABORATORY EXPERIMENT MANUAL



EDITED BY ASSOC. PROF. DR. ALEV ER

01- ATOMIC SPECTRA and DETERMINATION of RYDBERG'S CONSTANT

PURPOSE of the EXPERIMENT

1. To investigate the relationship between atomic spectra and the energy levels of atoms.

2. To find the Rydberg constant.

GENERAL INFORMATION

With the help of the radiation emitted and absorbed by atoms, it is possible to learn about their structure. Many theories and models have emerged due to the studies carried out to understand the structures of atoms. One is the "Thomson Atomic Model" of J.J. THOMSON (1856-1940). According to this model, Thomson designed the atom as electrons (-) distributed in a positively (+) charged sphere. He thought that the placement of these electrons in the sphere should be such as to make the atom uncharged. After Thomson, New Zealand physicist Ernest RUTHERFORD (1871-1937) proposed the "Rutherford Atomic Model". According to this model, he proved by experiments that the positive charges are collected in the center of the sphere called the "Nucleus", not in the sphere. Following the results of these experiments, he likened the atom and electrons to the solar system; he designed the (+) charged nucleus instead of the sun and electrons instead of planets. The rotational motion is realized by "Centripetal Force", in other words, it is an accelerated motion. According to the electromagnetic theory, charges in accelerated motion will give out light, so the electron's energy will decrease and fall into the nucleus. This is not the case.

The "Bohr Atomic Model" developed by the Danish physicist Niels BOHR (1825-1962) after both models are still valid for the Hydrogen atom today. According to this model, Bohr put forward two postulates:

- The electron can only travel in circular orbits whose angular momentum is an integer multiple of $\hbar$. $(mvR = n\hbar)$

A hydrogen atom can exist in any one of certain energy states (with specific discrete energy) for a very short time, called the lifetime of the level. The lowest-energy state of the atom is called the "ground state" and the other higher-energy states are called "excited states".

- An electron emits or absorbs electromagnetic waves as it passes from one orbit to another.

Only if the hydrogen atom passes from one fixed state, i.e. energy level, to another, does it emit or absorb radiation with an energy equal to the difference between the energies of these two states. $(E_2 - E_1 = h\nu)$

Absorption: The passage of an atom from one energy level to a higher energy level by taking energy.

Spontaneous emission: The radiation of an atom that has passed to a higher energy level with an energy equal to the difference between the two energy levels as it passes back to the lower energy level.



Absorption Spontaneous emission

Figure 1: Absorption and Emission.

The energy of any steady state of the hydrogen atom is given by the expression;

$$E_n = -\frac{1}{8}\frac{e^4 m_e}{\varepsilon_0^2 h^2}\frac{1}{n^2} \qquad n = 1,2,3 \ldots..$$

Here "n" is the " Principal Quantum Number".

For the H atom;

$$E_n = -13{,}6(1/n^2)\ eV \quad n = 1,2,3..$$

will be in the form. Figure 2 shows the Energy-Level diagram determined for the H atom.

The wavelength of the spectral transition can be calculated as follows:

$$E_{n_1} = -\left(\frac{me^4}{8\varepsilon_0^2 h^2}\right)\frac{1}{n_1^2}, \qquad E_{n_2} = -\left(\frac{me^4}{8\varepsilon_0^2 h^2}\right)\frac{1}{n_2^2} \qquad \Delta E = E_{n_2} - E_{n_1}$$

$$E_{n_1} = -\left(\frac{me^4}{8\varepsilon_0^2 h}\right)\left(\frac{1}{n^2} - \frac{1}{n}\right) = h\nu, \qquad h\nu = \frac{hc}{\lambda} = \left(\frac{me^4}{8\varepsilon_0^2 h}\right)\left(\frac{1}{n^2} - \frac{1}{n}\right)$$

$$\frac{1}{\lambda} = \left(\frac{me^4}{8\varepsilon_0^2 h^2}\right)\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

$$\frac{1}{\lambda} = R_{th}\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

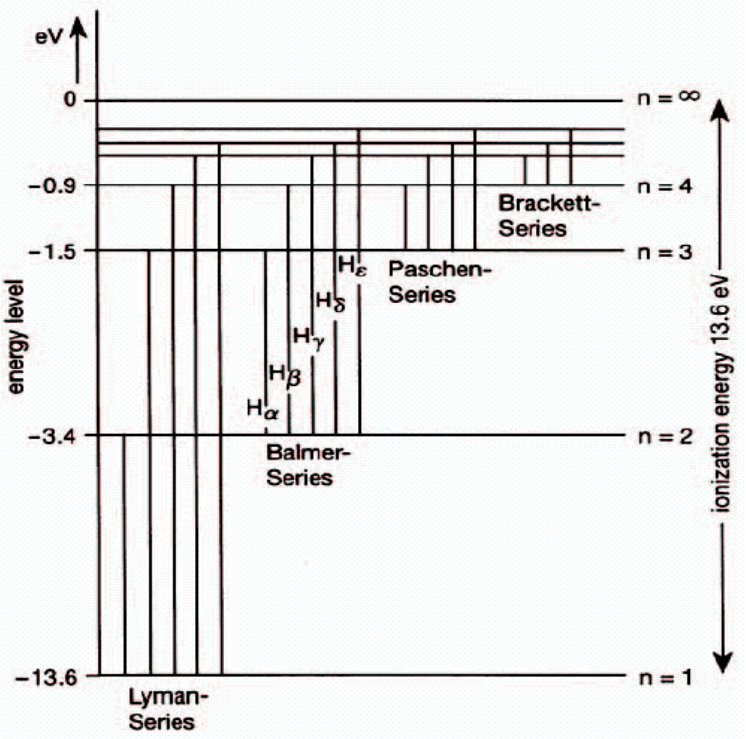
Where n_2 and n_1 are the principal quantum numbers corresponding to the upper and lower energy levels respectively.

λ is the "Wavelength" of the emitted photon.

Rydberg Constant:

$$R_H = \frac{1}{8}\frac{e^4 m_e}{\varepsilon_0^2 h^3 c} = 1.097.10^7 \text{ m}^{-1}$$

1/λ : Wave number in (m-1) dimension.



$n = 1$: Lyman series
Spectral range: ultraviolet
$n = 2$: Balmer series
Spectral range: ultraviolet till red
$n = 3$: Paschen series
Spectral range: infrared
$n = 4$: Bracket series
Spectral range: infrared
$n = 5$: Pfund series
Spectral range: infrared

Figure 2: Energy levels of the Hydrogen Atom.

HOW TO CONDUCT THE EXPERIMENT

The experimental setup to be used in this experiment is shown in Figure 3. Tools used in the experiment: Grating, Hydrogen and Mercury lamp, Power Supply, Ruler.



Figure 3: Atomic Spectra and determınatıon of Rydberg's constant experimental setup.

Figure 4 shows schematically the diffraction of light in the grating.

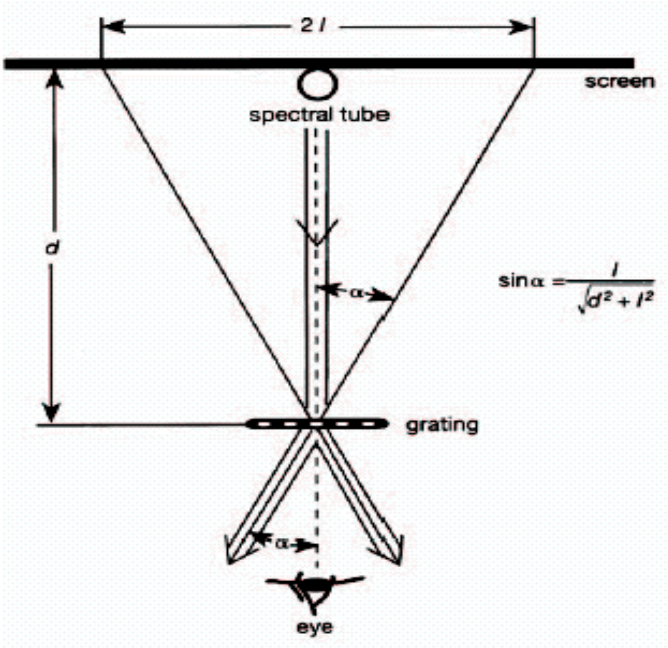


Figure 4: Grating

If light of wavelength λ is incident on a grating with a grating constant g, it diffracts. Maximum intensity is observed if the following condition is met.

$$m\,\lambda = g \sin\alpha \qquad m = 0,1,2,3$$

For nth-order diffraction, the following equation applies.

$$m\,\lambda = g\frac{l}{\sqrt{d^2 + l^2}} \qquad g = 1.671\mu m$$

A hydrogen lamp is placed in the experimental setup. The constant of the diffraction net used in the experiment is $g = 1.671\mu m$

1. Observe the colors on the ruler.
2. For each color, identify the same colors on the ruler on either side of the central maximum and read the length $2l$ on the ruler (see Figure 4).
3. Measure the distance d between the grating and the ruler.
4. Calculate the wavelength of the color you see using the formula below. Here, if you are reading the colors in the first row, take m=1.

$$m\,\lambda = g\frac{l}{\sqrt{d^2 + l^2}} \qquad g = 1.671\mu m$$

5. Using Figure 2, determine the numbers n_1 and n_2 for the lines H_α, H_β, H_γ. Fill in Table 1 by determining the R_{th} Rydberg constant (theoric for the lines H_α, H_β, H_γ of the Balmer series with the help of the formula below using the wavelengths you found in point 4 and the n numbers you determined here.

$$\frac{1}{\lambda} = R_{th}\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

6. Do the same for the Hg lamp and fill in Table 2 according to the colors you observed.

Table 1: Results

Line	**2*l* (m)**	**n_1**	**n_2**	**λ_{exp}**	**λ_{lit}**	**R_{exp}**
H_α					656.28 nm	
H_β					486.13 nm	
H_γ					434.05 nm	
R_{ave}=..... ±						

Table 2: Results

Hydrogen		Mercury	
Measured Wavelength (nm)	**Reference Value (nm)**	**Measured Wavelength (nm)**	**Reference Value (nm)**
	656.28		579.067
	486.13		546.075
	434.05		491.6068
	410.17		435.8335

QUESTIONS ABOUT THE EXPERIMENT

1. What is a "spectrum"? Explain "Absorption and Emission Spectra".
2. Explain "Binding Energy" and "Ionization Energy".
3. Explain the events that form the basis for the formation of the "Rainbow".

REFERENCES

1. H Haken, H C Wolf, The Physics of Atoms and Quanta, 2000, Springer
2. http://hyperphysics.phy-astr.gsu.edu/HBASE/quantum/atspect.html
3. Phywe Experimental literature and operating manuals

02-ELECTRON DIFFRACTION

PURPOSE of the EXPERIMENT

To observe the diffraction of free electrons through a polycrystalline graphite layer and prove that electrons have wave characteristics. To calculate the lattice constants d_1 and d_2 of graphite.

GENERAL INFORMATION

In 1923, the French physicist L. de Broglie thought massive particles should also have wave behavior. According to Lois De Broglie's hypothesis, any particle with momentum p, h=6.626.10-34 js Planck's constant

$$\lambda = \frac{h}{p} \quad (1)$$

It also has the character of a wave with a wavelength. In 1927, Davisson and Germer experimentally confirmed the De Broglie hypothesis by showing that electrons diffract from nickel crystals. De Broglie was awarded the Nobel Prize in Physics in 1929.

The de Broglie wavelength of an electron accelerated by the voltage V_a can be found;

$$KE = \frac{p^2}{2m} = eV_a$$

$$p = \sqrt{2emV_a} \quad \Rightarrow \quad \lambda = \frac{h}{\sqrt{2emV_a}} \quad (2)$$

An opaque obstacle, a slit, or a diffraction network can be used to observe the diffraction phenomenon. In this experiment, a diffraction grating obtained by sputtering carbon on a Ni (Nickel) grid was used as a target. 12 mg C contains 6.1023 atoms. Bragg diffraction occurs when waves are scattered from crystalline solids. The diffraction of electron waves sent onto the crystal structure is shown in Figure 1.

$$AC = d \sin\theta$$

$$ACB = 2d \sin\theta$$

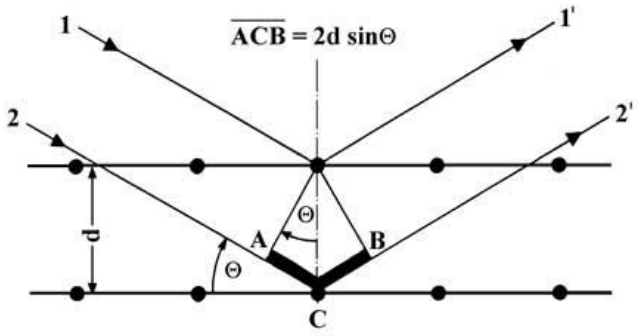


Figure 1: Brag diffraction

Waves incident on a crystalline solid can be scattered from different planes in the crystal structure. This scattering from different layers causes a path difference between the waves. The path difference between waves scattered from neighboring planes is given by $2d\sin\theta$, where d is the distance between the planes. Bragg diffraction condition for the angle θ that the reflected wave makes with the surface of the solid,

$$2d \sin\theta = n\lambda \quad ; \quad n = 0,1,2,... \quad (3)$$

If the electron behaves as a wave, a diffraction pattern should be observed on a screen L away from the crystal. Indeed, as can be seen in this experiment, two rings with a bright center are observed on the screen. The change in the anode voltage (Va), which accelerates the electrons, causes the radius to change. In other words, as the voltage decreases, the ring radius value increases diffraction increases. This result is in agreement with de Broglie's assumption. A decrease in voltage leads to an increase in wavelength, which leads to an increase in angular dispersion due to diffraction conditions.

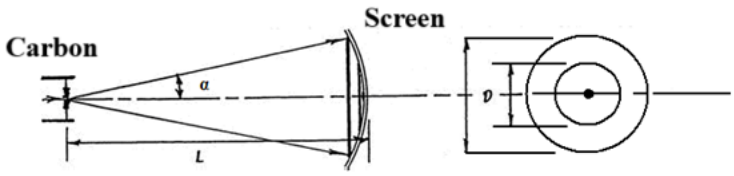


Figure 2: The geometric shape of the interference pattern.

Electron waves from the crystal are scattered at an angle of 2θ with the direction of incidence of the electrons.

$$\tan 2\theta = \frac{R}{2L}$$

When we take into account the small-angle approach,

$$\lambda = d\frac{R}{2L}$$

$$R = \frac{2Lh}{d\sqrt{2meV_a}} \qquad (4)$$

The radius of the glass bulb used in the experiment is 6.75 cm.

$$d = \frac{2Lh}{R\sqrt{2meV_a}} \qquad (5)$$

Graphite crystals have a hexagonal geometry. Therefore, electron waves sent to graphite crystals will undergo Bragg diffraction from two different spaced planes. The width of these intervals is d_1= 0.213 nm and d_2= 0.123 nm as given in Figure 3.

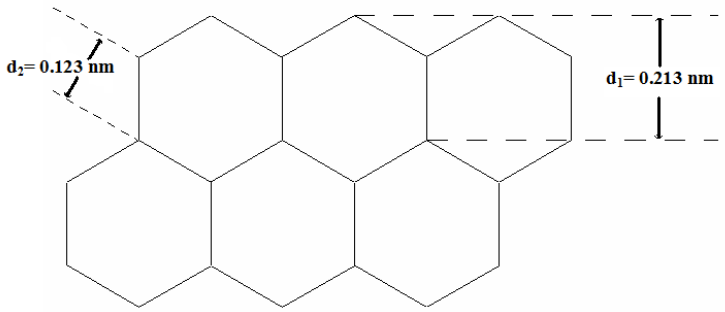


Figure 3: Crystal structure of graphite.

In this case, for Bragg diffraction consisting of two separate gaps, two separate rings with diameters R_1 and R_2 will be formed. The crystal gaps caused by these rings are as follows.

$$d_{10} = \frac{2Lh}{R_1\sqrt{2meV_a}} \qquad d_{11} = \frac{2Lh}{R_2\sqrt{2meV_a}} \qquad (6)$$

Here *m* and *e* are the mass and charge of the electron.

HOW TO CONDUCT THE EXPERIMENT

Electron diffraction tube, high voltage source (0-10 kV), power supply (0-600 V)

The electron diffraction experimental setup in which diffraction will be observed is shown in Figure 4.

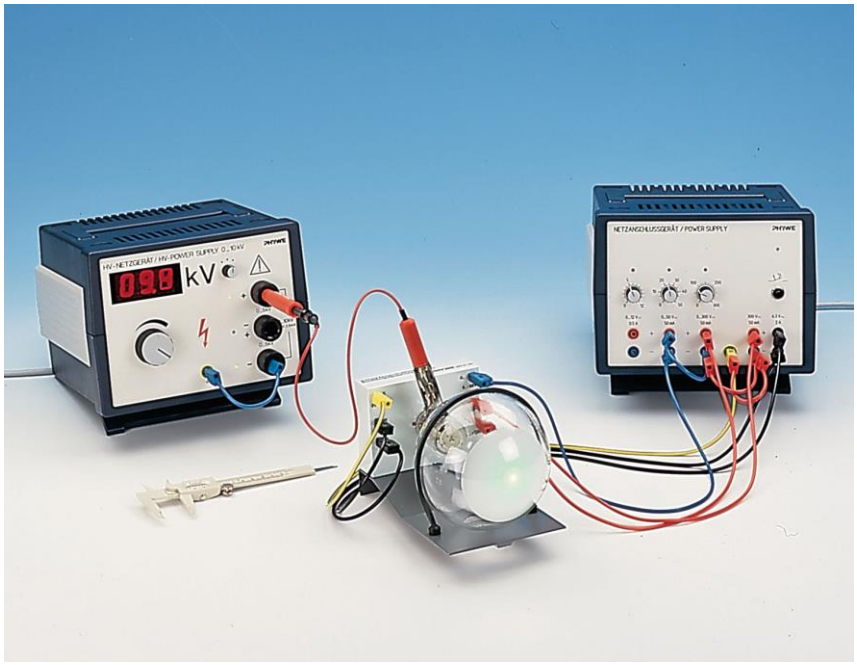


Figure 4: Experimental setup

The experimental setup consists of an electron diffraction tube and a high-voltage power supply. In the diffraction tube, electrons released from the electron gun are accelerated and pass through the polycrystalline graphite sample. Some of the accelerated electrons undergo Bragg diffraction from these crystals. The diffracted electrons produce a diffraction pattern on the fluorescence screen at the front of the tube.

1. Apply a voltage of 6V to the heater circuit. Wait 2 minutes for the circuit to fully warm up.
2. Apply a high voltage of 5kV to accelerate the electrons.
3. For the diffraction pattern you see on the screen of two luminous concentric rings, vary the voltage, measure the diameter of the rings for each voltage value, and record them in the table.
4. Calculate d_{10} and d_{11} from the values in the table and calculate the average value with error.

5. Plot $\frac{1}{\sqrt{V_0}}$-graphs separately with the diameter values of R_1 and R_2 found in step 3. Calculate the crystal spacings d_{10} and d_{11} by substituting the slope you found from these graphs in relation (6). Compare with the values found in step 4.

Table 1: Results

V (kV)	R_1(cm)	R_2(cm)	d_{10}	d_{11}

REFERENCES

1. H Haken, H C Wolf, The Physics of Atoms and Quanta, 2000, Springer
2. Phywe Experimental literature and operating manuals
3. http://www.cce.ufes.br/jair/web/difr_el.pdf

03-ELEMENTARY CHARGE and MILLIKAN EXPERIMENT

PURPOSE of the EXPERIMENT

This experiment involves observing charged oil droplets accelerated between two capacitor plates.

1. Measurement of oil droplets' rise and fall times with various charges at different voltages.

2. Determination of the charge and radius of oil droplets.

3. Determination of the elementary charge from the velocity of the oil droplets in the direction of gravity and the opposite direction.

4. Determination of the e/m ratio of the electron.

GENERAL INFORMATION

The increasing and decreasing motion of charged oil droplets in the electric field of the capacitor is observed and their velocity is determined:

The velocity of the fall of the oil droplet in the electric field	υ_1
The velocity of the rise of the oil droplet in the electric field	υ_2
The voltage of the capacitor	U
Charge on droplets	$Q = n.e$
The radius of droplets	r
Capacitor inner-electrode distance	$d = 2.5mm \pm 0.01mm$
Density of silicone oil	$\rho_1 = 1.03.10^3 kgm^{-3}$
Viscosity of air	$\eta = 1.82.10^{-5} kg(m.s)^{-1}$
Acceleration due to gravity	$g = 9.81ms^{-2}$
Density of air	$\rho_2 = 1.293kgm^{-3}$

The force F applied to a sphere of radius r and velocity υ in a viscous, non-fluid liquid of viscosity η:

$$F = 6\,\pi\, r\, \eta\, \upsilon \qquad \text{(Stoke Kanunu).} \qquad (1)$$

A droplet of mass m, volume V, and density ρ_1 is located in the earth's gravitational field.

$$F = m.g = \rho_1.V.g \qquad (2)$$

Buoyancy is given as follows:

$$F = \rho_2 .V.g \qquad V = \frac{4}{3}\pi\, r^3 \tag{3}$$

Electric field force,

$$F = Q.E = Q.\frac{U}{d} \tag{4}$$

From the total forces acting on the charged particle, the velocities of the fall and rise of the droplets are obtained.

$$\upsilon_1 = \frac{1}{6\pi r\eta}\left(Q.\frac{U}{d} + \frac{4}{3}\pi r^3 g(\rho_1 - \rho_2)\right) \tag{5}$$

$$\upsilon_2 = \frac{1}{6\pi r\eta}\left(Q.\frac{U}{d} - \frac{4}{3}\pi r^3 g(\rho_1 - \rho_2)\right) \tag{6}$$

Adding and subtracting these equations gives the charge and radius of the droplet.

$$Q = C_1 \frac{\upsilon_1 + \upsilon_2}{U}\sqrt{|\upsilon_1 - \upsilon_2|} \tag{7}$$

$$C_1 = \frac{9}{2}\pi d\sqrt{\frac{\eta^3}{g(\rho_1 - \rho_2)}}$$

$$C_1 = 2.73.10^{-11} kgm(m.s)^{-1/2}$$

$$r = C_2\sqrt{|\upsilon_1 - \upsilon_2|} \tag{8}$$

$$C_2 = \frac{3}{2}\sqrt{\frac{\eta}{g(\rho_1 - \rho_2)}}$$

$$C_2 = 6.37.10^{-5}(m.s)^{1/2}$$

Figure 1 shows that the charge of the droplets has precise values whose coefficient is the elementary charge *e*.

$$Q = n.e$$

The average value is obtained as the elementary charge $e = 1.68.10^{-19}$ As.

$$kg\sqrt{\frac{m}{s}}\frac{\left(\frac{\mathrm{m}}{\mathrm{s}}\right)}{\mathrm{V}}\sqrt{\frac{\mathrm{m}}{\mathrm{s}}}=kg\frac{m}{s}\frac{\left(\frac{\mathrm{m}}{\mathrm{s}}\right)}{\left(\frac{\mathrm{kg\,m^2}}{\mathrm{s^3\,A}}\right)}=\mathrm{kg}\frac{\mathrm{m}^2}{s^2}\frac{\mathrm{s}^3A}{kg\,\mathrm{m}^2}=As=C$$

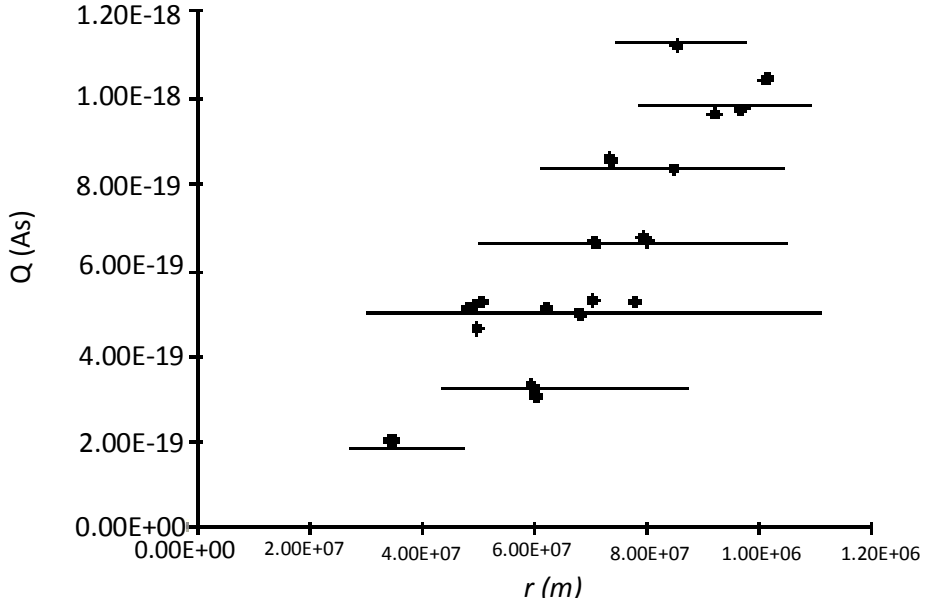


Figure 1: Various droplets measured by the Millikan Method to determine the elementary charge.

HOW TO CONDUCT THE EXPERIMENT

The experimental setup is as follows.

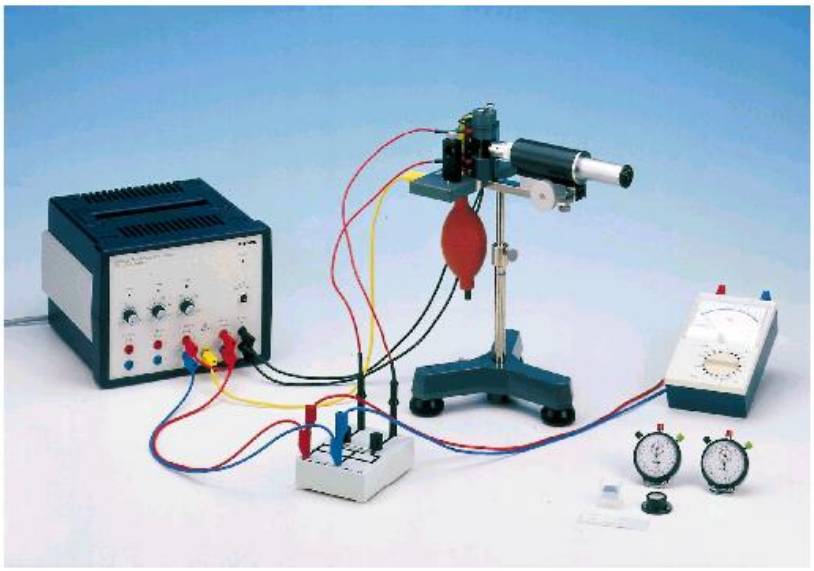


Figure 2: Experimental setup for the determination of the elementary charge with Millikan oil drop method.

Multi-range meter, Power Supply (0...600 V DC), Stage micrometer (1 mm-100 divisions), Stopwatch, Coated Glasses (18318 mm) 50 pcs, Commutator Switch, Stand Tube, Connection Cables.

The microscope's eyepiece micrometer is calibrated with a stage micrometer. The stage micrometer has two small rings in the center and the center of the smallest one is a scale. This scale is used for eyepiece micrometer adjustment.

1. To do this, a stage micrometer is slid into the slits between the microscope lens and the open small chambers (Figure 3).

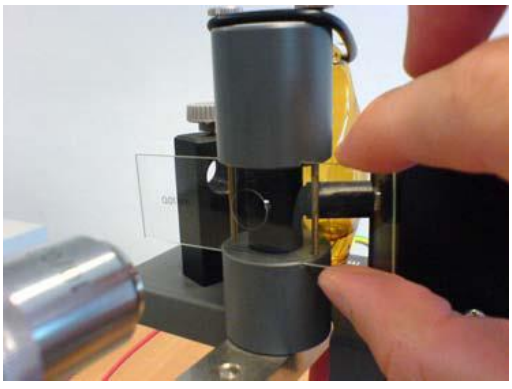


Figure 3

2. Look through the microscope and adjust the focus so that the scale is visible (the two rings around the scale should be used as orientation). The distance between the microscope lens and the stage micrometer should be only a few millimeters.

3. The two scales should overlap.

4. Now count the lines of the stage micrometer, the eyepiece micrometer should be covered by 30 lines.

5. Therefore, convert the 30 divisions of the eyepiece micrometer into mm so that 100 lines on the stage micrometer correspond to 1 mm.

6. Record the calibration result.

7. Remove the stage micrometer from the slits to start the experiment.

8. Sometimes it is useful to put a cover glass in the slits to protect the chamber against drafts.

9. Set the multi-range meter to the 600 V D.C. measurement region.

10. Set the capacitor voltage to a value between 300 V and 500 V. (Set the rotary switch for voltage on the power supply to 0.)

11. Look into the microscope and move some oil droplets with the sprayer several times.

12. Adjust the focus of the microscope until oil droplets like small white rings appear (Figure 4).

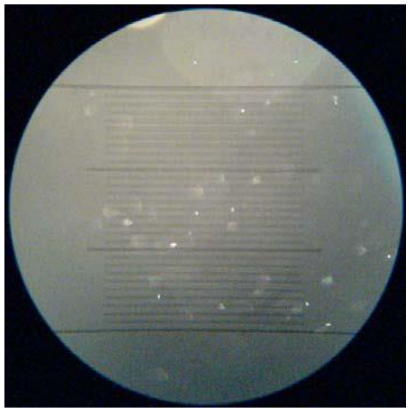


Figure 4

13. Observe the movement of the oil droplet and find its charge. (An oil droplet can be identified as a charge when the polarity of the capacitor is reversed and the orientation is changed by the commander switch setting.)

14. To see the oil droplet for a long time, the microscope must be properly focused.

15. Try moving the charged oil droplet several times between the highest and lowest degrees on the eyepiece micrometer with the setting of the commander switch.

16. When you do not see any oil droplets, spray some droplets again (sometimes it is useful to turn the power supply off and then on again after a few seconds).

The experiment should be carried out by at least two people.

1. Select an oil droplet and use the commutator switch to move the droplet between the highest and lowest degrees on the eyepiece micrometer. Correct the focusing of the microscope if necessary.

Note: Criteria to follow when selecting oil droplets:

- The droplet should not move too fast, it should have a small charge (between 1 and 3 s for 30 divisions).
- The droplet should not move too slowly and should not show excessive movements. If desired, the capacitor voltage is increased.

2. As soon as you find the oil droplet, collect a few ascent times with the first stopwatch (t_1) and a few descent times with the second stopwatch (t_2). Notice that the image you see under the microscope is inverted. Write the results in Table 1.

Note: For both cases the addition times must be greater than 5 seconds.

3. With the help of the calibration, convert the distances traveled by the oil droplets into "s" mm and write your results in Table 1.(columns 4 and 7)

4. Repeat the measurement twice, 400 V and 500 V.

Velocity values υ_1 and υ_2 for each of the three oil droplets

$$v = \frac{s}{t}$$

With the above equation, s_1 and t_1 or s_2 and t_2 can be calculated from the measurement results shown in Table 1. Do not forget to convert the s_1 and s_2 results into m units. Write your results below in Table 2.

5. Now write down your results in Table 2 using the equations below and calculate the charges Q along with the radii r of the oil droplets. Find the mean of the result and calculate its error.

$$Q = C_1 \frac{v_1 + v_2}{U} \sqrt{|v_1 - v_2|}$$

$$C_1 = 2.73x10^{-11} kg(m/s)^{-1/2}$$

$$r = C_2 \sqrt{|v_1 - v_2|}$$

$$C_2 = 6.37x10^{-5}(mxs)^{1/2}$$

6. Combine the possible Q charge and r radius results for the oil droplets to decide the elementary charge *e* from the measurement results. Plot the Q charges corresponding to the radius r of the oil droplets (as in Figure 1). Interpret the graph.

7. Now draw the lines defining each charge level. The graph should look like the one in Figure 1.

8. In the next step, determine the distance between each line. These distances correspond to the elementary charge, *e*. Remember the distance between the x-axis (Q=0) and the first line. Calculate the distances and average them.

Calibration 30 divisions = 0.89 mm (to be calculated by calibration)

Table 1: Results

Voltage U[V]	Ascent times t_1[s]	Distance traveled s_1[div]	Distance traveled s_1[mm]	Descent times t_2[s]	Distance traveled s_2[div]	Distance traveled s_2[mm]
300						
300						
300						
400						
400						
400						
500						
500						
500						

Table 2: Results

Voltage U[V]	Ascent velocity v_1 [m/s]	Descent velocity v_2 [m/s]	Charge Q [C]	Radius *r* [m]
300				
300				
300				
400				
400				
400				
500				
500				
500				
Average Q:				

9. Plot the charges Q corresponding to the radius r of the oil droplets.

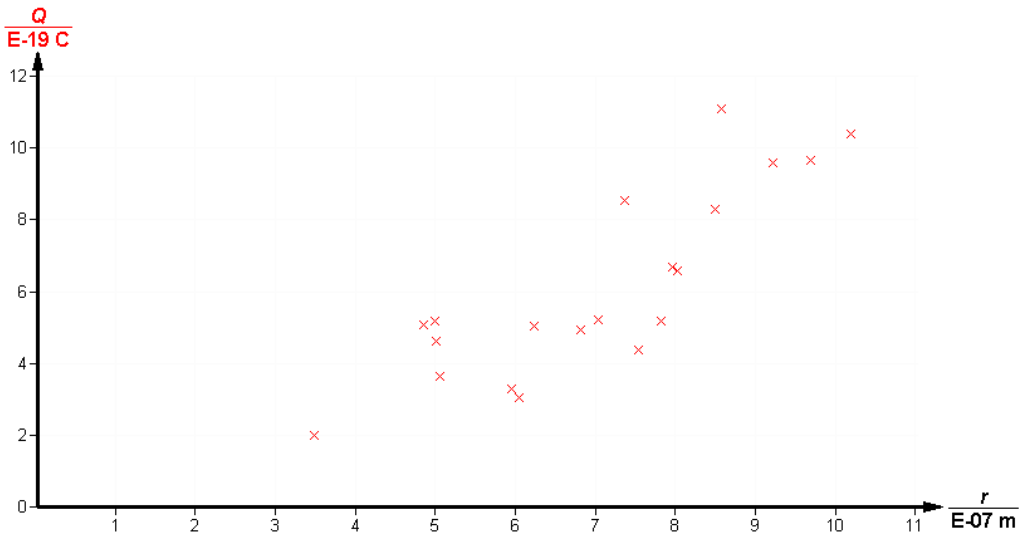


Figure 5: Plot of the Q charges corresponding to the radius r of the oil droplets

10. Draw the lines defining each charge level. The graph should look like Figure 6.

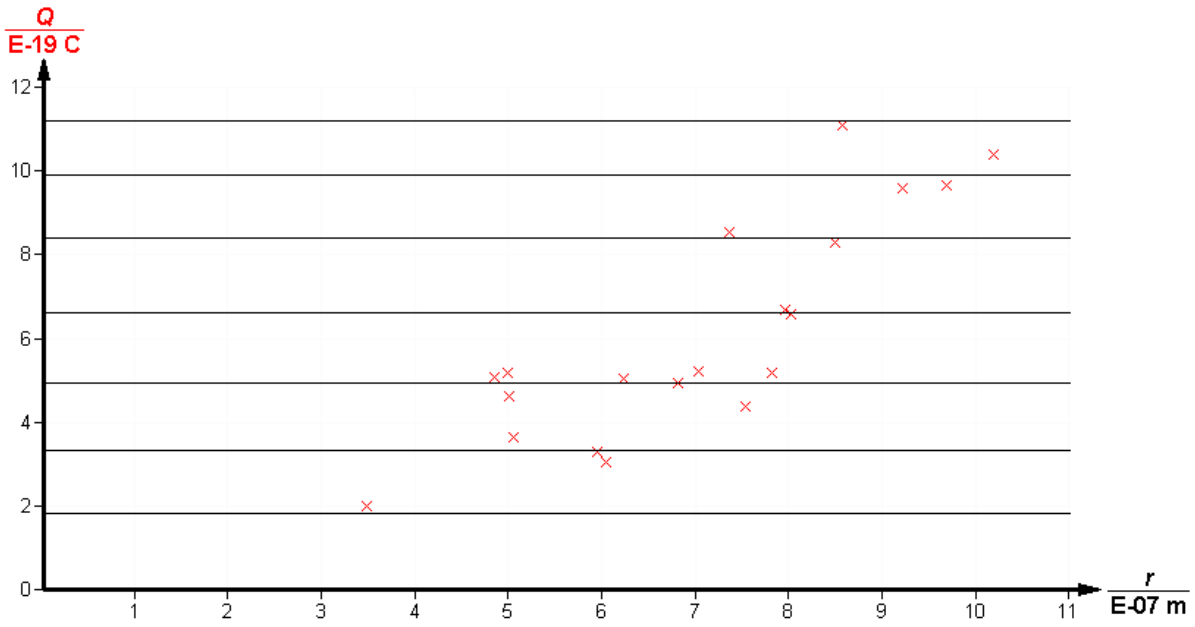


Figure 6

11. Determine the distance between each line. These distances correspond to the elementary charge, *e*. Remember the distance between the x-axis (Q = 0) and the first line. Calculate the distances and tabulate the results in Table 3.

Table 3: Results

Investigated lines	**Difference [C]**
x-axis and first line	
First and second line	
Second and third line	
Third and fourth line	
Fourth and fifth line	
Fifth and sixth line	
Sixth and seventh line	
Average value	

The value of the elementary charge *e* in the literature is $e = 1.602 \times 10^{-19}$ C.

Questions:

1. Explain why charged oil droplets rise or fall.

2. In this experiment several forces act on the charged oil droplet. Draw two graphs: the first graph to show the forces acting on the charged oil droplet as it rises, and the second graph to show the forces acting on the charged oil droplet as it falls.

3. What relation can be observed between the elementary charge and the charge of an oil droplet? Given that N is the number of charges, *e* is the basic charge and Q is the charge of the oil droplet, formulate a formula for the relationship between them.

REFERENCES

1. Catalog of Phywe Elementary Charge and Millikan Experiment
2. www.aip.org

04-DETERMINATION of the e/m RATIO of the ELECTRON

PURPOSE of the EXPERIMENT

Determination of the e/m ratio of the electron by studying the motion of cathode rays accelerated under different electric potentials in a uniform magnetic field.

GENERAL INFORMATION

If a free electron with charge e moves with a velocity v in a vacuum perpendicular to a magnetic field B, the magnetic force acting on this charge is given by the following relation.

$$\overrightarrow{F_B} = e\vec{v}x\vec{B} \qquad (1)$$

In the experimental setup, electrons released by heating the cathode are accelerated under a voltage V and sent between Helmholtz coils. Since there is no magnetic field as long as no current flows through the Helmholtz coils, the electron beam continues on its way without any deflection.

The electrical energy of the electrons with charge e accelerated by the accelerator voltage V is converted into the kinetic energy of the electron eV.

$$eV = \frac{1}{2}mv^2 \qquad (2)$$

This gives the following expression.

$$v = \left(\frac{2eV}{m}\right)^{1/2} \qquad (3)$$

A magnetic field is created between the coils by passing a current between 0-1A through Helmholtz coils. As the current is increased, a uniform magnetic field B is created in the region between the coils. The strength of the magnetic field is set so that it is much larger than the force of gravity so that the force of gravity is ignored. The magnetic force exerted on the electron by the magnetic field is perpendicular to the electron's velocity and forces the electron

into a circular motion. This magnetic force provides the centripetal force necessary to move in a circular orbit.

$$m\frac{v^2}{r} = evB \qquad (4)$$

If v is substituted here and the e/m ratio is created, the following expression is obtained.

$$\frac{e}{m} = \frac{2V}{B^2r^2} \qquad (5)$$

As can be seen from this expression, the ratio of the electron's charge to its mass is determined by the variation of the accelerating voltage concerning the magnetic field for a circular orbit of constant radius.

HOW TO CONDUCT THE EXPERIMENT

The experimental setup is as follows.



Figure 1: Experimental setup

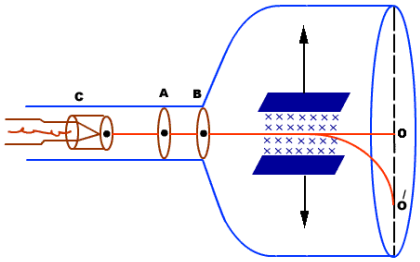
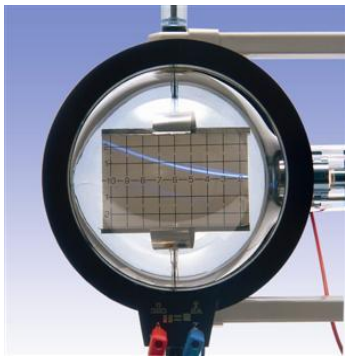
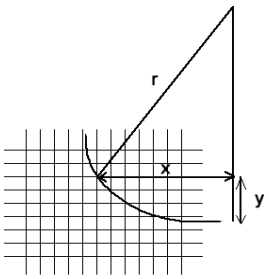


Figure 2: Cathode ray tube

Determination of r

The radius of the circular orbit of the electron beam on the fluorescent screen can be calculated from the figure:





$$r^2 = x^2 + (r - y)^2$$

$$r = (x^2 + y^2)/2y \qquad (6)$$

Figure 3: On-screen image of the electron trajectory.

Determination of B

One method for obtaining a uniform magnetic field is to use a pair of Helmholtz coils with equal currents flowing through them and with a common axis. For a coil of a given radius, the spacing between the coils must be equal to the radius to obtain the optimum uniform field.

The magnetic field between the coils is given by the Biot-Savart law, as follows,

$$B = \frac{\mu_0 N R^2}{(R^2 + a^2)^{3/2}} I \qquad \mu_0 = 1 \qquad (7)$$

N: number of turns on each coil (given as 320)

R: radius of the coil (6.8 cm)

a: average distance between coils (6.8 cm)

I: the amount of current flowing through each coil.

Thus the relation between current and magnetic field is given by $B = 4{,}25.10^{-3}\ m^{-1}\ I$ when the constants are substituted.

As the current increases, the magnetic field strength increases accordingly. In the experiment, the e/m_e ratio is calculated by measuring V, I, x, and y values.

V (Volt)	**I (A)**	**r (cm)**	**B (Tesla)**	***e*/*m*$_e$ (C/kg)**
Average *e*/*m*$_e$ =				

REFERENCES

1. http://hyperphysics.phy-astr.gsu.edu/hbase/magnetic/helmholtz.html
2. Phywe experimental literature and operating manuals

05-ELECTRON SPIN RESONANCE (ESR)

PURPOSE of the EXPERIMENT

This experiment aims to investigate the magnetic moment of the electron spin.

1. Determination of the Landé-g factor of the DPPH (Diphenylpicrylhydrazyl) sample.

2. Determination of the half-width of the absorption line (FWHM).

GENERAL INFORMATION

Electron spin resonance (ESR) is based on the energy absorption by substances with unpaired electrons in an external magnetic field generated by a DC source.

Determination of Landé-g Factor

There are two different atomic magnetic dipoles. Circular currents represent the circulation of electrons around the atomic nucleus, and magnetic moments depend on the electron spin.

The magnetic moment generated by a circular current,

$$\vec{A} = A.\vec{e} = \pi.R^2.\vec{e}$$

Here A is the surface of the region where the electron moves,

R, orbital radius,

$\vec{e}$ is the unit vector perpendicular to the surface.

ω, is the angular velocity, $T = 2\pi / \omega$ is the period of the orbit, and the orbital magnetic moment is

$$\vec{\mu}_{orbital} = I.\vec{A} = -\frac{e}{T}.\pi.R^2.\vec{e} = -\frac{e.\omega}{2}.R^2.\vec{e} \qquad (1)$$

given by the equation.

The magnetic moment is directly proportional to the angular momentum $\vec{L}$. m_e is the mass of the electron;

$$\vec{L} = m_e.\vec{r}x\vec{v} = m_e.\omega.R^2.\vec{e} \qquad (2)$$

given by the equation.

If this expression is substituted in equation (1),

$$\vec{\mu}_{orbital} = -\frac{e}{2.m_e}.g_{orbital}.\vec{L} = \gamma_{orbital}.g_{orbital}.\vec{L} \qquad (3)$$

is obtained. Here, $g_{orbital}$ is the "Landé Factor" and $\gamma_{orbital} = -\frac{e}{2.m_e}$ is called the "Gyromagnetic Ratio". A comparison of equations (1) and (2) shows that the momentum of a pure orbital Landé-factor $g_{orbital}$ = 1 is obtained.

$$\vec{\mu}_{orbital} = \gamma_{orbital}.\vec{L} \qquad (4)$$

The spin angular momentum $\vec{S}$ represents the spin of an electron. The electron's magnetic moment is directly proportional to this spin angular moment.

$$\vec{\mu}_{spin} = -\frac{e}{2.m_e}.g_{spin}.\vec{S} = \gamma_{spin}.\vec{S} \qquad (5)$$

$$\gamma_{spin} = -g_{spin}.\frac{e}{2.m_e} = g_{spin}.\gamma_{orbital}$$

One purpose of this experiment is to obtain the Landé-g coefficient for the electron spin. Just as electrons have spins, nuclei also have spins. However since the mass of the nucleus is much larger than the mass of the electrons, $\gamma_{çekirdek}$ is very small. Therefore the magnetic moment of the nucleus can be neglected.

According to quantum mechanics, angular momentum takes discrete values.

$$L_Z = m\hbar \qquad (m = l, l\text{-}1, l\text{-}2, \ldots..1\text{-}l, l) \qquad (6)$$

Where *m* is the magnetic quantum number, h=6.63.10^{-34} Js is Planck's constant and $\hbar \equiv \frac{h}{2\pi}$.

The numerical value of the spin angular momentum can only be 1/2.

$$S_Z = \pm\frac{1}{2}\hbar \qquad (7)$$

Magnetic moments are also quantized and defined in terms of the Bohr magneton.

$$\mu_B = \frac{e}{2.m_e}.\hbar = 9.27x10^{-24}\,Am^2 \qquad (8)$$

z-components of orbital and spin angular momenta are,

$$\mu_{z,orbital} = \gamma_{orbital}.L_z = -m.\mu_B \qquad (m=l, l\text{-}1, l\text{-}2, \ldots.1\text{-}l, l) \qquad (9)$$

$$\mu_{z,spin} = \gamma_{spin}.S_z = g_{spin}.\gamma_{orbital}.S_Z = \left(\mp\frac{1}{2}\right).g_{spin}.\mu_B \qquad (10)$$

given by these equations.

Total angular momentum if an electron has both orbital angular momentum and spin angular momentum,

$$\vec{J} = \vec{L} + \vec{S} \text{ ve } |J| = |L-S|, |L-S+1|, |L-S+2|, ..., |L+S|$$

The corresponding total magnetic moment,

$$\vec{\mu}_j = \vec{\mu}_l + \vec{\mu}_s = -\frac{e}{2.m_e}.\vec{L} - \frac{e}{2.m_e}.g_{spin}.\vec{S} = -\frac{\mu_B}{\hbar}.\left(\vec{L} + g_{spin}.\vec{S}\right) \qquad (11)$$

If L and S values are known, the Landé-g coefficient can be calculated as follows:

$$g_j = 1 + \frac{J.(J+1) + S.(S+1) - L.(L+1)}{2.J.(J+1)} \qquad (12)$$

The DPPH sample used in the experiment has one unpaired electron. Its orbital magnetic moment $L = 0$ and the total magnetic moment consists only of the spin magnetic moment and is $g_{spin} = g_j$. This value is approximately equal to the Landé factor of a free electron. Substituting $S = 1/2$ and $L = 0$ in equation (12) gives $g_j = 2$. However, the actual value of g is slightly larger than 2 due to other interactions that depend on B.

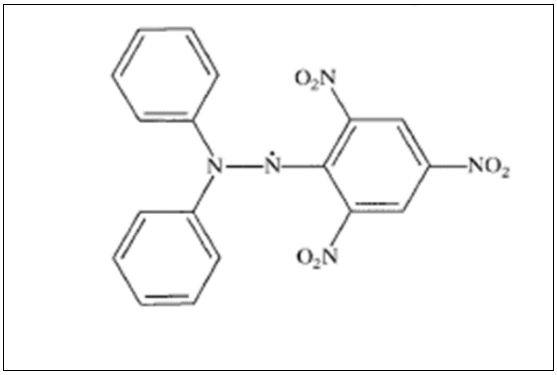


Figure 1: DPPH (Diphenylpicrylhydrazyl) sample

Zeeman Effect and Magnetic Resonance

The Zeeman Effect describes the splitting of the atomic spectral line due to an external magnetic field applied to an atom or molecule. This effect is caused by the interaction between the

magnetic moments of the atom or molecule and the applied external magnetic field. The following expression applies to the potential energy in a magnetic field:

$$E = -\mu.B \tag{13}$$

There are two types of Zeeman effects, Normal Zeeman ($S = 0$) and Anomalous Zeeman ($S \neq 0$). Since only the orbital angular momentum (L) exists in the normal Zeeman effect, only the orbital magnetic moment and the external magnetic field interact. When the external magnetic field is applied, the energy levels of the atom split into equally spaced (ΔE) levels:

$$\begin{aligned} \Delta E &= -\Delta\mu_{orbital}.B \\ &= \frac{e}{2.m_e}\Delta L_z.B \\ &= \mu_B.\Delta m.B \end{aligned} \tag{14}$$

But as in this experiment, the electron spin must also be taken into account. Thus, the magnetic field interacts not only with the orbital magnetic moment but also with the spin magnetic moment. In this case, the Zeeman interaction is written in the following form:

$$\Delta E = \mu_B.\Delta m_j.g_j.B \tag{15}$$

With the selection rule $\Delta m_j = \mp 1$ for magnetic transitions, the distance between two Zeeman levels is as follows:

$$\Delta E = \mu_B.g_j.B \tag{16}$$

For the transition from a low energy level to a high energy level to take place, energy must be absorbed whose absolute value is equal to the energy difference between the two levels. This energy comes from the electromagnetic wave of frequency f provided by the ESR resonator. This energy,

$$E = hf \tag{17}$$

This process is called “Magnetic Resonance”.

The resonance condition is achieved by varying the B_r magnetic field by changing the voltage applied to the Helmholtz coils during the experiment.

The Landé factor is calculated according to the following equation:

$$\mu_B.g_j.B_r = hf$$

$$g_j = \frac{h.f}{\mu_B.B_r} \tag{18}$$

Here; h = 6.63.10^{-34} Js, f = 146 MHz, μ_B = 9.27.10^{-24} Am^2, if these values are substituted;

$$g_j = 10{,}43.10^{-3} T.\frac{1}{B_r} \tag{19}$$

is obtained. Here T is the abbreviation for tesla.

The magnetic field in the ESR experimental set used is provided by a pair of Helmholtz coils of turns ω = 241 and radius R = 4.8 cm.

$$B = \mu_o.\frac{8.I.\omega}{\sqrt{125.R}} = 0.7155.\mu_o.\frac{I.\omega}{R} \tag{20}$$

Here μ_0 = 4π10^{-7} Tm/A and I is the current through the coil. If the values are substituted;

$$B = 0.6445.\mu_o.\frac{I.\omega}{R}$$

$$B = 4.07.10^{-3} I \ \frac{T}{A} \tag{21}$$

is obtained. The value of g is calculated by substituting the resonant current I_r into equation (22):

$$g = \frac{2.565A}{I_r} \tag{22}$$

HOW TO CONDUCT THE EXPERIMENT

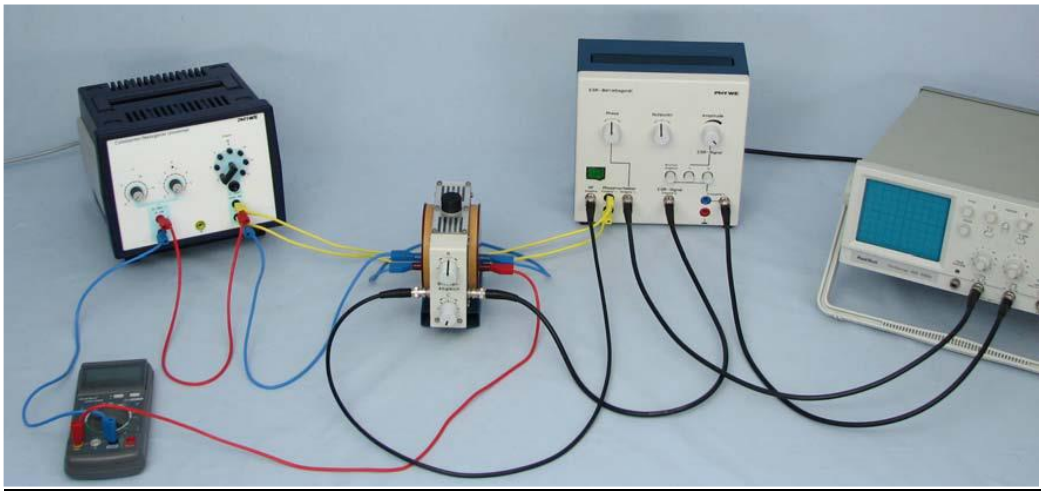


Figure 2: ESR experimental setup

Determination of Landé-g Factor

The ESR test setup is shown in Figure 2.

1. Make sure that the direct voltage (DC) switch labeled "V" on the main power unit is set to "0" and the current switch labeled "A" is set to "5A".

2. The alternating voltage must be set to 2V (this corresponds to a frequency of 50 Hz).

3. It is important that the direct voltage is overlapped by the alternating voltage so that the ESR signal is displayed on the oscilloscope.

4. Then switch on the main power unit, the ESR power unit and the oscilloscope.

5. Press the "Bridge Balancing" button (button 8) on the ESR power unit (Figure 3).

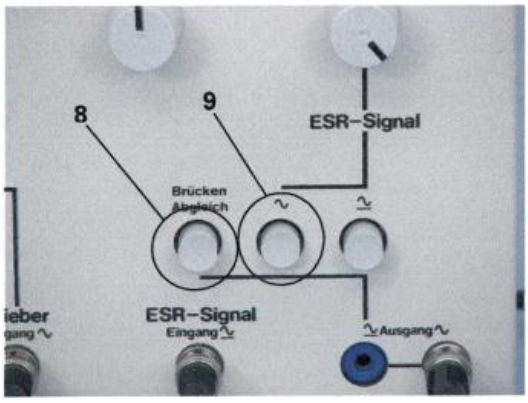


Figure 3: Part of the ESR power unit

6. The "R" switch of the resonator should be in the center position and the "C" switch in the far left position.

7. On the oscilloscope, select the "X-Y" mode (Figure 4).



Figure 4: "X-Y" mode

8. Select "GND" mode for channel "X" and "d.c." mode for channel "Y".

9. The signal sensitivity for both channels should be 1 V/cm (Figure 5).



Figure 5: The signal sensitivity

10. A single point should now be visible on the oscilloscope.

11. Move the "Position" switch to the point displayed in the center of the coordinate system.

12. Next, press button "9" on the ESR power unit (Figure 3) and select the d.c. mode for channel "X" on the oscilloscope.

13. You should now see a horizontal line on the oscilloscope.

14. Increase the direct voltage on the main power unit until the digital multimeter reads 1.3 A.

15. Now carefully turn the "C" knob on the resonator to the right until a signal appears on the oscilloscope. (It may be important to increase the intensities of the X and Y channels to 0.5 V/cm or more to obtain a stronger signal.)

16. As soon as the signal appears, the two lines coincide with the "Phase" button of the ESR power unit.

17. Adjust the signal with "C" until you see a symmetrical shape with a minimum. (Try to make the signal as symmetrical as possible.)

18. The minimum point in the figure should be brought exactly to the y-axis by reducing the direct voltage on the main power unit. (Use "C" again to get a symmetrical figure.)

19. If the figure is as follows (Figure 6), you have obtained a good "resonant signal".

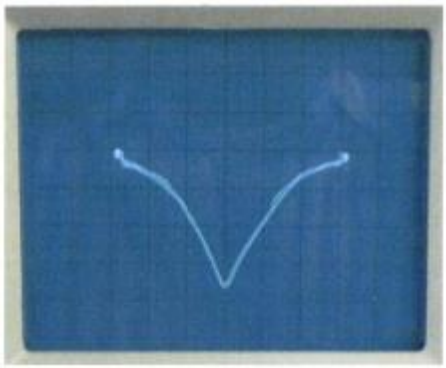


Figure 6: The resonant signal

20. The current value read from the multimeter at the moment when this resonant signal is obtained is called "Resonant Current" (I_r).

21. Record this I_r current in Table 1. Repeat all the steps so far two more times and record the I_r currents you have found in Table 1.

22. Substitute these current values in the equation $g = \frac{2.565A}{I_r}$ and calculate the value of g with error and record it in Table 1. The value in the literature is g = 2.0037.

Table 1: Results

I_{r1}	g_1	I_{r2}	g_2	I_{r3}	g_3
			g_{ave} =	±	

Determination of FWHM (Full Width at Half Maximum)

1. Use the "Position" switches to adjust the position of the signal so that the x-axis passes exactly half the height of the signal.

2. Starting from the zero point, count the scaled divisions until the signal and x-axis coincide. The more symmetrical the signal, the more precise and accurate this measurement will be).

3. The number of these scaled divisions depends on the sensitivity you have chosen. Therefore, make sure that you do not change the signal sensitivity during the rest of the experiment.

4. Record your results below.

5. Intersection (RIGHT SIDE) =

6. Intersection (LEFT SIDE) =

7. To determine the half-width (FWHM), the distance between the two intersections must be measured in Amperes.

8. Therefore, the alternating voltage must be switched off and the resonator must be connected directly to the direct voltage. For this purpose, the red wire connected to the direct voltage input with the alternating voltage input must be disconnected and the blue wire of the resonator must be connected to the now empty direct voltage input (Figure 7).

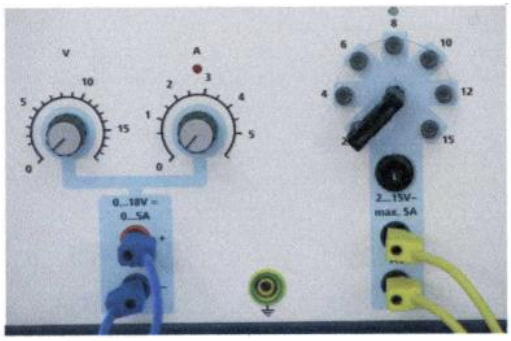


Figure 7: Part of the power supply

9. Disconnect the BNC cable from the ESR power unit connected to the "X" channel of the oscilloscope.

10. Connect the appropriate adapter to the end of this BNC cable as shown in Figure 8.



Figure 8: BNC cable

11. Then connect this adapter to the direct voltage inputs of the main power unit (Figure 9).



Figure 9: The direct voltage inputs

12. Make sure that the sensitivity of the "X" and "Y" channels is not changed during this experiment.

13. Vary the correct voltage until only a single dot appears on the oscilloscope.

14. When the dot appears, use the "position" knob to change its position until the dot is on the x-axis.

15. Move the point you have found to one of the two intersections you found earlier by changing the correct voltage on the main power unit.

16. Note the current value (I_1) read from the multimeter at this moment.

$$I_1 =$$

17. Now move the point to the other intersection and note the second current (I_2).

$$I_2 =$$

Calculate the absolute value of the difference between the currents I_1 and I_2 to obtain the width at half maximum. Write down your results below.

$$\Delta I = \left| I_1 - I_2 \right|$$

18. Then calculate the magnetic displacement by writing the value of ΔI in equation (21) and note your result below. This value you found is the FWHM. The value in the literature is $\Delta B = 2{,}8.10^{-4}$ Tesla.

$$\Delta B = \ldots\ldots\ldots\ldots$$

REFERENCES

1. http://www.nikhef.nl/~h73/kn1c/praktikum/phywe/LEP/Experim/5_1_12.pdf
2. Phywe experimental literature and operating manuals

06- FINE STRUCTURE of TWO-ELECTRON ATOM He

PURPOSE of the EXPERIMENT

Determination of the wavelengths of the spectral lines of the two-electron helium (He) atom

GENERAL INFORMATION

In the ground state of the He atom, two s electrons fill the innermost shell with a principal quantum number $n = 1$. In the excited state, one electron remains in the half-filled first shell while the other is excited to the upper shell. Thus

1. Electron $l = 0$ in the case of $n = 1$ and

2. $l = 0...n\text{-}1$ if electron $n > 1$

has two electrons.

The term diagram (Grotrian diagram) for the helium atom is given in Figure 1. This diagram shows the allowed electronic transitions between the energy levels of atoms. They can be used for single-electron and multi-electron atoms and are constructed considering the selection rules. Helium in the singlet state is also called parahelium. Unlike alkali atoms, it has no fine structure. All its lines are single. The symbol 1^1S gives the lowest term. Here the first 1 is the head quantum number, the higher index 1 is the multiplicity (singlet) and the letter S stands for the total orbital angular momentum, which is zero. The larger terms are 2^1S, 2^1P, 3^1S, 3^1P, 3^1P, 3^1D, etc. Due to the lack of fine structure, they are oriented in such a way that the antiparallel and vectorial sum of the spins of the two electrons gives a total spin of $S = 0$. Unlike the single system, triplet helium has a fine structure and is called orthohelium. Its lowest level is 2^3S. Here 2 stands for the excited electron with $n = 2$, the top index 3 stands for the multiplicity term and the letter S stands for $L = 0$.

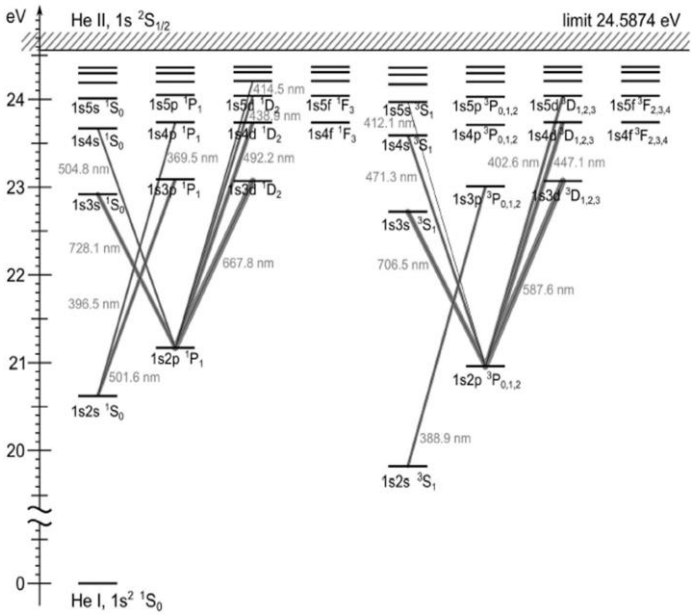


Figure 1: Term diagram of the He atom.

In this system, the spins of the two electrons are parallel to each other. Quantum number of total spin $s_1 + s_2 = S = 1$. Combinations between systems cannot be observed, i.e. there are no optical transitions between singlet and triplet systems. If one compares the mutual configurations in two non-combining systems, one finds that there are noticeable differences in energy, especially for low quantum numbers. The 2^1S_o state is 0.8 eV above the 2^3S_1 state and the 2^1P_1 state is around 0.25 eV above the 2^3P_2 state. This energy difference arises from the difference in electrostatic interactions between two electrons with parallel and antiparallel spin orientations.

The energy eigenvalues E in the atomic system can be found using the time-independent Schrödinger equation:

$$H\Psi(r_i) = E\Psi(r_i) \qquad (1)$$

Here Ψ is the wave function, r_i is the position of the i. electron, and H is the Hamiltonian energy operator and represents the total energy of the system. In many electron systems, this equation cannot be solved exactly because the interactions between electrons are not spherically symmetric. Some approximations are made to solve the equation.

When the spectra of many-electron atoms are analyzed, it is understood that the lines thought to be single lines are not single. As a result of the interaction between the spin and orbital moment of the electrons and the internal magnetic field, splits in the energy levels of the atom, called fine structure, occur. Electrons have an orbital angular momentum (L) together with an

emergent magnetic moment (μ_L) and a spin angular momentum (S) together with a magnetic moment (μ_S). The electron moves in the electric field of the nucleus and therefore has a magnetic field proportional to its angular momentum l. The orientational energy of the electron in the field is E_{nl}:

$$E_{nl} = -\boldsymbol{\mu}_S.\mathbf{B} = \zeta\, \boldsymbol{s}.\boldsymbol{l} \qquad (2)$$

ζ is the fine structure constant.

In many electron systems, the Hamiltonian is written as follows:

$$H = \sum_{i=1}^{N}\left(-\frac{\hbar^2}{2m}\nabla_i^2 - \frac{Ze^2}{(4\pi\varepsilon_0)r_i}\right) + \sum_{j>i=1}^{N}\frac{e^2}{(4\pi\varepsilon_0)r_{ij}} + \sum_{i=1}^{N}\zeta_i(r_i)\,\vec{s}_i\cdot\vec{\ell}_i \qquad (3)$$

The first term in this equation gives the kinetic energy of the electrons, the second term gives the Coulomb interaction between the nucleus and each electron, the third term gives the Coulomb interaction between electrons and the last term gives the spin-orbit interaction. If the interaction between the outer electrons is greater than the interaction of the spin and orbital moments for each electron, the LS Coupling (Russell-Saunders Coupling) is observed.

LS Coupling (Russell-Saunders Coupling)

The interaction between each angular momentum must be taken into account. The vector sum of spin and orbital angular momentum is given in the figure below

$$\boldsymbol{L} = \sum \boldsymbol{l}_i \text{ ve } \boldsymbol{S} = \sum \boldsymbol{s}_i \qquad (4)$$

is obtained. Values that the total angular momentum quantum number can take,

$$\boldsymbol{J} = \boldsymbol{L} + \boldsymbol{S}, \qquad (5)$$

The combination of the electron configuration of an atom and the term symbol defines the energy eigenvalues of the atom. The quantum number L determines the term characteristics:

L = 0, 1, 2,.... corresponds to the terms

$S, P, D, \ldots$

When determining terms in LS coupling,

$$n^{2S+1}L_J \qquad (6)$$

Where n is the principal quantum number and $2S+1$ is the multiplicity term. The quantum number J takes the value $(2L+1)$ if $L < S$ and $(2S+1)$ if $L > S$. Each value of J corresponds to an energy level characterized by the possible orientations of the L and S vectors. Due to the mutual interactions of the total spin and orbital angular momenta of the outer electrons, the terms are decomposed into subterms, called multiplets, defined by the total J quantum number. Taking

the helium atom as an example, let us show that atomic terms can be derived from given electron configurations.

If both electrons are in the lowest shell, the electron configuration is 1s1s or $1s^2$ in the usual notation. In this case, they have the following quantum numbers:

$$n_1 = n_2 = 1, \quad l_1 = l_2 = 0, \quad s_1 = 1/2, \quad s_2 = 1/2$$

The resulting quantum numbers of the atom are then either ^1S_o for the singlet ground state

$$L = 0, S = 0 \qquad m_{s1} = - m_{s2}, \qquad J = 0$$

or $^3S_1$ for the triple ground state

$$L = 0, S = 1 \qquad m_{sl} = m_{s2}, \qquad J = 1$$

will be of the form. In reality, only a single ground state is observed. The triple ground state is forbidden by Pauli's principle since two electrons would not only have the same n and l quantum numbers but also the same m_s spin orientation. However, if one electron remains in the shell with $n = 1$ and the other is promoted to the state with $n = 2$ (1s2s electron configuration), we have the following quantum numbers:

$$n_1 = 1, n_2 = 2, \; l_1 = l_2 = 0, s_1 = 1/2, s_2 = 1/2$$

This gives either the singlet state $L = 0$, $S = 0$, $J = 0$, i.e. ^1S_o, or the triplet state $L = 0$, $S = 1$, $J = 1$, i.e. $^3S_1$. Both states are allowed and observed. In the same way, states and term symbols can be derived for all electron configurations.

Selection Rules

For a multi-electron system, the following selection rules apply:

$$\Delta L = 0, \pm 1$$

$$\Delta S = 0$$

$$\Delta J = 0, \pm 1 \qquad (J = 0 \rightarrow J = 0 \text{ Forbidden})$$

The spectral lines in Figure 1 are shown in Table 1 together with the term values of the lower and upper energy levels.

Table 1: Spectral lines of the He atom.

Color	λ/nm	Relative Transition	Relative Intensity
Red	706.5	$3\ ^3S \rightarrow 2\ ^1P$	5
Red	667.8	$3\ ^1D \rightarrow 2\ ^1P$	6
Red	656.0	He-II	4 – 6
Yellow	587.6	$3\ ^3D \rightarrow 2\ ^3P$	10
Green	504.8	$4\ ^1S \rightarrow 2\ ^1P$	2
Green	492.2	$4\ ^1D \rightarrow 2\ ^1P$	4
Blue	471.3	$4\ ^3S \rightarrow 2\ ^3P$	3
Blue	447.1	$4\ ^3D \rightarrow 2\ ^3P$	6
Blue	438.8	$5\ ^1D \rightarrow 2\ ^1P$	3
Purple	414.4	$6\ ^1D \rightarrow 2\ ^1P$	2
Purple	412.1	$5\ ^3S \rightarrow 2\ ^3P$	3
Purple	402.6	$5\ ^3D \rightarrow 2\ ^3P$	5
Purple	396.5	$4\ ^1P \rightarrow 2\ ^1S$	4
Purple	388.9	$3\ ^3P \rightarrow 2\ ^3S$	10

HOW TO CONDUCT THE EXPERIMENT



Figure 2: Experimental set for the calculation of the spectral lines of the helium atom.

As shown in Figure 2, the spectrometer consists of four main parts. These are the collimator, the goniometer, the area where the grating is placed, and the binoculars. The collimator has an adjustable slit spacing and parallelizes the beams.

If light of wavelength λ falls on a grating with a diffraction constant d, it is diffracted. Maximum intensities are found when the diffraction angle θ satisfies the following conditions for m = 0, 1, 2...:

$$m\,\lambda = d\,sin\theta$$

m is the order number. The value of the grating constant used in this experiment is d = 1684 nm.

In this experiment, the emission lines of a He lamp having two electrons will be analyzed. The excitation of He atoms with two electrons in the lamp media is made possible by electron collisions. The energy difference resulting from the excited state E_1 to the ground state E_o is emitted as a photon with frequency f:

$$hf = E_1 - E_0$$

Here h is Planck's constant ($h = 6.63.10^{-34}$ J.s).

1. Light the He lamp. The wavelengths of the visible colors of the He atom are given in Table 2.

Table 2: The spectral lines of the He atom that can be observed during the experiment whose wavelengths will be found, and their term values.

Color	λ/nm	Relative Transition	Relative Intensity
Red	667.8	$3\ ^1D \rightarrow 2\ ^1P$	6
Yellow	587.6	$3\ ^3D \rightarrow 2\ ^3P$	10
Green	504.8	$4\ ^1S \rightarrow 2\ ^1P$	2
Green	492.2	$4\ ^1D \rightarrow 2\ ^1P$	4
Blue	471.3	$4\ ^3S \rightarrow 2\ ^3P$	3
Blue	447.1	$4\ ^3D \rightarrow 2\ ^3P$	6

2. Place the grating on the table and adjust the table so that the light is perpendicular to it.

3. Set the binoculars to the zero bright line and read the angle on the goniometer.

4. Then turn the binoculars to the left and then to the right and read the values of θ_{right} and θ_{left} for the colors given in Table 2 at m = 1 and m = 2.

5. Subtract the zero-order angle you read in Step 3 from the diffraction angles θ_{left} and θ_{right} and enter them in Table 3.

6. For each spectral line in each color, calculate the average of the diffraction angles you found on the left and right sides.

7. Calculate the λ(nm) wavelengths of the colors you see using the formula $m\lambda = d\ (sin\theta_{ave})$.

8. Calculate the average of the wavelengths for each color found at $m = 1$ and $m = 2$ and record them in Table 3.

9. Draw the calibration curve of the spectrometer for the values of λ and $sin\theta_{ave}$ determined for $m = 1$.

Table 3: Results

m = 1

λ	θ_o	θ_{left}	$(\theta_{left}-\theta_o)$	θ_{right}	$(\theta_{right}-\theta_o)$	θ_{ave}	$sin\theta_{ave}$	λ(nm)
λ_{red}								
λ_{yellow}								
λ_{green}								
λ_{blue}								
λ_{purple}								

m = 2

λ	θ_o	θ_{left}	$(\theta_{left}-\theta_o)$	θ_{right}	$(\theta_{right}-\theta_o)$	θ_{ave}	$sin\theta_{ave}$	λ(nm)
λ_{red}								
λ_{yellow}								
λ_{green}								
λ_{blue}								
λ_{purple}								

	***m* = 1**	***m* = 2**	λ_{ave} (nm)
λ_{red}			
λ_{yellow}			
λ_{green}			
λ_{blue}			
λ_{purple}			

REFERENCES

1. H Haken, H C Wolf, The Physics of Atoms and Quanta, 2000, Springer
2. Phywe experimental literature and operating manuals
3. http://hyperphysics.phy-astr.gsu.edu
4. Svanberg, S., 2001, Atomic and molecular spectroscopy: Basic aspects and practical applications, Springer

07-PHOTOELECTRIC EFFECT

PURPOSE of the EXPERIMENT

1. Explanation of the photoelectric effect by observing it experimentally.

2. Finding Planck's constant (*h*) with the help of the photoelectric effect.

3. Representation of the maximum kinetic energy of photoelectrons as a function of the frequency of light.

4. Calculation of the work function and threshold frequency of the metal at the photocathode.

GENERAL INFORMATION

The photoelectric effect was first observed by H. Hertz in 1887. While Hertz was experimenting with electromagnetic waves, he noticed that the electric arcs formed in the air gap between the cathode and the anode formed more easily when ultraviolet light was sent to the cathode. Hertz himself did not dwell on this observation. However other physicists tried to understand this effect. It soon became clear that the cause of this effect was the emission of electrons from the cathode when the frequency of light incident on the cathode was high enough. Thus, light had the effect of removing electrons from a metal surface. This phenomenon is called the "photoelectric effect". The electrons removed by light are called "photoelectrons".

The fact that light has the effect of removing electrons from a metal surface is a phenomenon that can be explained by the classical theory of electromagnetic waves. For this, it is sufficient to consider that electromagnetic waves are composed of electric and magnetic fields oscillating in directions perpendicular to each other.

The electric field of an electromagnetic wave exerts a uniform force on a charged particle, the electron. Here, $\vec{E}$ is the electric field and e is the charge of the electron. Due to the repulsion caused by this force, an electron can be removed from a metal surface. For this reason, the photoelectric phenomenon did not initially surprise physicists and was thought to be explainable by classical physics. However, more detailed experiments on the photoelectric phenomenon showed that it could not be explained by classical physics.

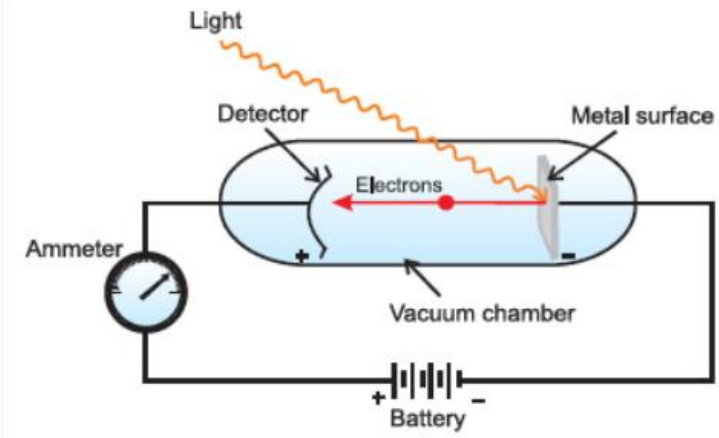


Figure 1: Photoelectric effect test tube

In 1902, P. E. A. Lenard conducted experiments to determine how the energy of photoelectrons removed from a metal plate by light depends on the intensity of the light incident on the plate. For this purpose, he illuminated a metal plate using a carbon arc lamp with adjustable light intensity.

He connected the collector plate to the cathode of a battery (Figure 1). The collector plate is thus negatively charged and a repulsion is created between the photoelectrons and the collector plate. Due to this stopping potential barrier, not all photoelectrons can easily reach the collector plate. Only those photoelectrons whose kinetic energy is large enough to overcome this stopping potential barrier can reach the collector plate. If the voltage applied by the battery is increased, no photoelectrons will reach the collector plate after a certain V value. This voltage V should be the maximum value of the kinetic energy of the photoelectrons. Thus, the photoelectrons reaching the collector plate generate a plate current which can be determined with the help of an ammeter.

$$eV_d = K_{max} \qquad \textbf{(1)}$$

According to the classical electromagnetic wave theory of light, as the intensity of light increases, the value of the electric field accelerating the electrons on the metal surface also increases. This means that the kinetic energy of the photoelectrons increases. This prediction is not consistent with the results of Lenard's experiments. Lenard repeated his experiment using light of a different color. He found that the kinetic energy of photoelectrons depends on the frequency of the light. When high-frequency light was used, the kinetic energy of the photoelectrons was also large.

The results of these experiments, which could not be explained by the classical electromagnetic theory of light, were announced by A. Einstein in 1905. Einstein predicted that the energy of light is not a continuous distribution of energy scattered along wavefronts, as predicted by the classical theory, but is concentrated in specific packets. In making this prediction, Einstein used the assumption that Planck had used to explain blackbody radiation. In 1900, to explain the nature of black body radiation, Planck proposed that the energies of the standing electromagnetic wave modes in a cavity could be analyzed as a function of their energy,

$$E_n = nh\nu \qquad \textbf{(2)}$$

Einstein assumed that Planck's assumption was valid for all electromagnetic waves, not just standing electromagnetic waves. According to Einstein's hypothesis, light is composed of quanta with energy $h\nu$. The number n indicates how many photons the beam contains and this number determines the intensity of the light beam. In this case, the energy of a single photon is determined only by its frequency. With this assumption, it is possible to explain the results of Lenard's experiment.

The American experimental physicist R.A. Millikan opposed Einstein's hypothesis because it contradicted the classical electromagnetic wave theory of light, but after 10 years of experimental work, he obtained results that confirmed Einstein's hypothesis, contrary to his initial expectations. Based on Einstein's assumption, Millikan succeeded in measuring Planck's constant with high precision. Millikan's experimental work on the photoelectric effect is one of the important studies that proved Einstein's hypothesis. These studies were considered by the Nobel committee as sufficient evidence to confirm Einstein's hypothesis about the photoelectric effect and Einstein was awarded the Nobel Prize in physics in 1921. Millikan also received the Nobel Prize in physics in 1923 for his experimental work on the photoelectric phenomenon and electric charge.

Einstein's hypothesis indeed successfully explains the experimental results obtained by Lenard and Millikan on the photoelectric phenomenon. Taking the energy of a photon as $h\nu$, the absorption of a photon by a metal surface increases the energy of an electron in the metal by $h\nu$. Electrons with increased energy cannot leave the metal surface immediately. Because there is a potential energy binding the electrons to the metal surface, it is necessary to do a work of Φ to separate the electron from the metal surface. Φ is the minimum energy required to detach an electron from a given metal surface. This energy is called the “work function”. When the

energy of the electron increases by $h\nu$, Φ of this energy must be spent to detach the electron from the metal.

Work function,

$$\Phi = h\nu_0 \qquad \textbf{(3)}$$

is given as ν_0 is the threshold frequency.

If $h\nu < \Phi$, there will be no electron removal, but if $h\nu > \Phi$, there will be removal, and the remaining energy $h\nu - \Phi$ will be the electron's kinetic energy. In this case the kinetic energy of the photoelectrons,

$$\frac{1}{2}m\mathrm{v}^2 = h\nu - \Phi \qquad \textbf{(4)}$$

can be written. As can be seen, the kinetic energy of the photoelectron is only linearly dependent on the frequency of the light. Photons with a frequency lower than this threshold frequency cannot remove electrons from the metal and the photoelectric effect does not occur. When the intensity of the light beam increases, it is only the number of photons contained in the beam that increases. There is no change in the energy of each photon. In this case, more photoelectrons will be removed from the metal surface, but the kinetic energy of these photoelectrons will not change.

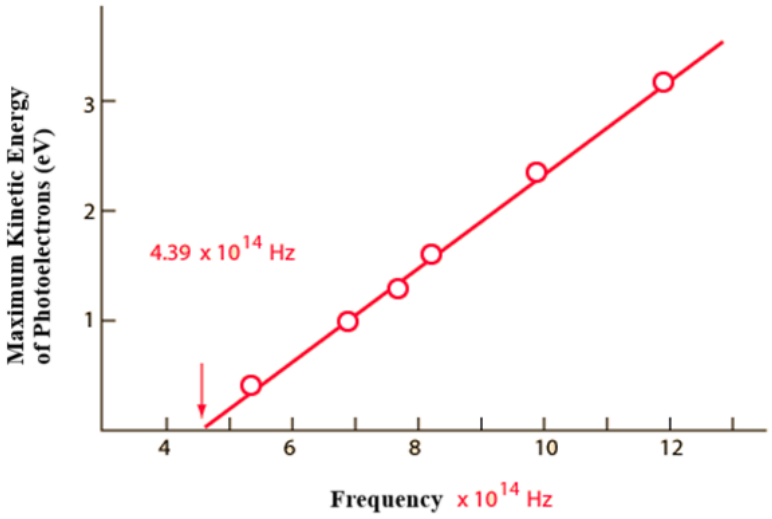


Figure 2: Frequency kinetic energy graph

It can be seen from relation (4) that the relationship between the kinetic energy of photoelectrons and the frequency of light is linear. If the kinetic energy of the photoelectron is plotted against the frequency of the photon, the graph is a linear line, the slope of the graph is Planck's constant and the point where the graph crosses the frequency axis gives the threshold frequency. Figure 2 shows the kinetic energy-frequency plot based on data obtained by Millikan in 1916.

$$eV_d = K_{max} = h\nu - \Phi \qquad \textbf{(5)}$$

From the slope of the graph, we find Planck's constant, $h = 4{,}16 \times 10^{-15}\, eV.s$. This value agrees with the value found by Planck with blackbody radiation. ($h = 4{,}1356675 \times 10^{-15}\, eV.s$ or $h = 6{,}626 \times 10^{-34}\, J.s$)

The findings of Hertz and subsequent experiments to explain the photoelectric phenomenon:

1. The photoelectric phenomenon is a physical phenomenon that supports the particle structure of light.

2. Polished metal surfaces illuminated with visible light and/or UV light emit electrons. These surfaces do not emit positive ions.

3. For these surfaces to emit electrons, a threshold energy (or a corresponding threshold frequency) must be exceeded.

4. The kinetic energy of the electrons (photoelectrons) detached from the surface does not depend on the light's intensity but on the light's frequency.

5. In the experiments, the time difference between the light hitting the metal surface and the electron detachment is ~10^{-9} s. This time is independent of the frequency of the light.

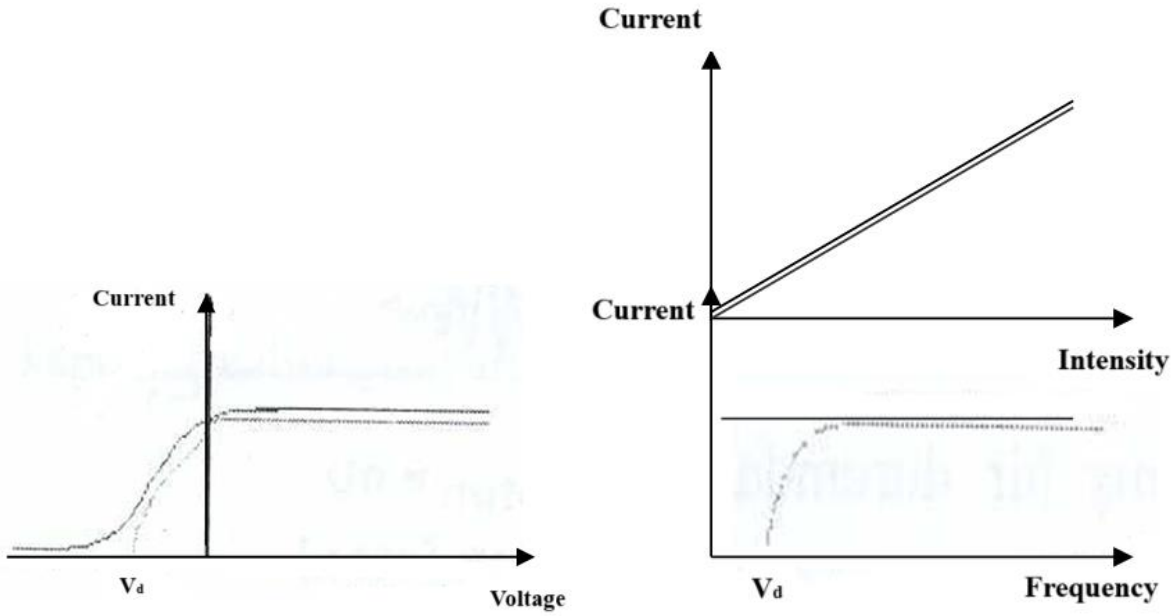


Figure 3: Variation of current with voltage, light intensity, and frequency of light

The current generated by photoelectrons depends on the light's intensity if the light's frequency is greater than a threshold value. As the intensity of the light increases, the current increases. In classical physics, the removal of electrons from a metal surface must occur at any frequency depending on the light intensity, whereas in the photoelectric event, electrons can be removed above a certain threshold value.

HOW TO CONDUCT THE EXPERIMENT

The experimental setup to be used is given in Figure 4.



Figure 4: Experimental set-up for determining Planck's constant (h) by photoelectric effect.

Elements in the optical setup shown in Figure 4:

1. Power Supply, **2.** Mercury Lamp (80 W), **3.** Slit (Adjustable), **4.** Convex Lens (f= +100 mm), **5.** Grating (600 lines/mm), **6.** Photocell, Electrometer (R>1013 Ω), **7.** Digital Multimeter, **8.** Filter (525 nm and 580 nm)

Steps

1. Turn on the Mercury Lamp and wait approximately 15 minutes for it to reach full brightness. If this lamp is turned off for any reason, it will need to cool down before it can light up again. Therefore, the lamp should not be turned off during the experiment.

2. Before starting the measurements, arrange the optical setup so that it is in the position shown on the goniometer.

3. A diffraction pattern appears on the screen. Rotate one arm of the optical assembly so that the entrance of the photocell overlaps the visible region spectral lines of the 1st order in the diffraction pattern on the screen. Read the angles at which each spectral line is observed with the help of the goniometer on the experimental setup and note them in Table 1.

$$m\lambda = d\sin\theta \qquad \textbf{(6)}$$

Using Bragg's law (relation (6)), calculate the wavelength and frequency values corresponding to each angle value and note them in Table 1. Check that each calculated wavelength corresponds to the spectral line spacing in Figure 5.

4. Move the optical setup over each spectral line and read the corresponding constant voltage values with a multimeter and note them in Table 1. (Wait for a few seconds to obtain constant and accurate voltage values).

5. To avoid 2nd-order UV diffraction phenomena, measurement values for the yellow and red spectral lines are obtained with the help of filters. Place a 525 nm wavelength filter for the yellow spectral line and a 580 nm wavelength filter for the red line at the inlet of the photocell read the corresponding constant voltage value and note it in Table 1.

6. Using the values obtained, form equation (5) for each color given in Table 1. Calculate 4 Planck constants by solving the possible binary equation systems obtained.

Example binary equation system:

$$eV_{d(violet)} = h\nu_{violet} - \Phi \qquad \textbf{(7)}$$

$$eV_{d(yellow)} = h\nu_{yellow} - \Phi$$

7. Draw the graph between the obtained stopping potential values and the frequency of light. Using the slope value in the graph, calculate Planck's constant and show that Planck's constant is compatible with the real value.

8. Calculate the metal's work function and threshold frequency in the photocathode.

Table 1: Results

Color	**Angle**	**Wavelength λ(nm)**	**Frequency ($\times 10^{14}$ Hz)**	**Stopping Potential (V)**
Violet				
Green				
Yellow				
Red				

$h_{experiment} = h \mp \Delta h$ J.s $\qquad$ $h_{graphic} = \ldots\ldots\ldots$ J.s

REFERENCES

1. Modern Fiziğin Kavramları Arthur Beiser
2. http://galileo.phys.virginia.edu/classes/252/photoelectric_effect.html
3. http://hyperphysics.phy-astr.gsu.edu/hbase/grexp.html
4. Phywe experimental literature and operating manuals

10- FRANCK-HERTZ EXPERIMENT

PURPOSE of the EXPERIMENT

The observation of the energy levels of atoms has discrete values. Recording the inverted current (inrush current) intensity I in a Franck-Hertz tube as a function of the anode voltage U. Calculation of the excitation energy E from the positions of the minima or maxima of the current intensity by difference formalization.

GENERAL INFORMATION

The explanation of critical physical phenomena of small particle systems was not possible with the classical mechanical view and this was the main reason for the emergence and establishment of the quantum mechanical view. Unfortunately, many attempts to determine the energy states of the atom using the classical view were unsuccessful, the common feature in all these failed attempts was that the classical view assumed the continuity of energy. In the early 1900s, Max Planck put forward a thesis that was not easy to accept, and which was met with a reaction from many scientific circles. Max Planck argued for the discontinuity of energy. I wonder if this idea would find the value it deserved at the time if it would break the stereotypes of a long period. It was indeed a way of thinking that was not easy to accept. However, it was not very difficult for this idea to be accepted, because many important problems in physics, which had not been solved until then, were perfectly solved within the framework of this view. Among these problems, the black body radiation solved by Planck has a very important place. In addition, Einstein, taking Planck's thesis into account, solved the photoelectric effect in 1905 and made a major contribution to the further consolidation of quantum theory.

J. Frank and G. Hertz were the first researchers to experimentally demonstrate the validity of quantum mechanical thinking and that atoms have discrete energy values. With their experiment on the mercury atom in 1914, these two scientists accomplished a very important feat that proved the validity of quantum theory.

Atoms can only have certain energy values, and these energy values are quantized (discrete) depending on the structure of the atom. These energy values, or in other words energy levels, are a distinctive physical characteristic of the atom. The energy levels above the fundamental energy level of the atom are called excited energy levels or excited energy levels. The energies of the energy levels of each atom are different from those of other atoms.

In our experiment, we aim to verify the above judgment by using the mercury atom. In this framework, our experiment aims to show that the atom will have discrete energy levels and to experimentally find only the first excitation energy of the mercury atom.

Atoms, the building blocks of matter, always prefer to be in the most stable state they can be in. The most stable state of the atom is the state in which the atom is at the fundamental energy level. In various ways, the atom passes from the stable state to a new state with external excitation, this state corresponds to one of the excited states of the atom. The atom is unstable in this excited state and wants to go back to the fundamental level, that is, the stable level. The time the atom can be in the excited state is a very small period, around 10-8 s, which is defined as the lifetime or lifetime of the excited level, after which the atom returns to the ground state. During this return, due to the conservation of energy, the atom emits a photon with an energy equal to the difference between the excited and ground state energies. This emitted photon can propagate at the speed of light in any direction of space.

Ei: Excited level energy and Es: ground-level energy

The energy of the emitted photon: $E_f = E_s - E_i = h\nu_f$

Here ν_f: frequency of the emitted photon, h: Planck's constant.

There are two different methods to excite an atom at the fundamental electronic energy level. Either accelerated electrons or ions are used to excite the atom or atoms are bombarded with photons. Both methods can excite the atom to the desired energy value, but there are significant differences between these two methods.

In this experiment, the mercury atom is excited with accelerated free electrons. When charged particles enter a uniform electric field, they gain potential energy from the electric field. The charged particles will convert this potential energy into kinetic energy during motion. If the charged particles manage to transfer their kinetic energy to the atoms through inelastic collisions with the atoms, they will perform excitation processes.

In the experiment, free electrons are obtained and accelerated in a uniform electric field to collide with mercury atoms in the gas phase. The conditions of the collisions are very important; the number of mercury atoms in the gas phase in the furnace where the collisions take place is an important factor for the probability of collisions and the mean free path, i.e. the distance between two consecutive collision points. In the experimental set, the conditions of the gas

environment were such that mercury atoms were excited only to the first excited energy level throughout the experiment.

In order to obtain a uniform electric field, it is sufficient to apply a direct voltage (D.C. potential difference) between the cathode and the anode. This voltage will be varied throughout the experiment with a potentiometer.

The cathode is heated with the help of a filament behind it to release electrons. These electrons are accelerated under the potential difference between the cathode and the anode. The potential energy of an electron under this accelerating voltage (V) is equal to the product of the electron's charge and the magnitude of the applied voltage.

Potential energy gained by free electrons: $E_p = eV$

During the movement starting from the cathode and continuing towards the anode, the electrons convert their potential energy into kinetic energy, as long as there is no energy loss, the kinetic energy they have when they reach the anode $E_k = eV = ½\ mv^2$

Interactions between free electrons and mercury atoms can take two forms; if the interactions take the form of collisions of the flexible type, the electrons collide with the mercury atoms like two billiard balls colliding with each other and continue their motion without any loss of kinetic energy. In this case, the atoms they collide with also remain at the fundamental electronic level without excitation. We call this type of interaction, which does not change the kinetic energy of the free electrons, elastic interactions. In this case, the electrons reach the anode with a kinetic energy of $E_k = eV$, depending on the voltage applied between the cathode and the anode.

The second possibility is in the form of inelastic collisions. In such a case, some of the free electrons may transfer their kinetic energy to the mercury atoms with which they collide, in which case the free electron loses all or most of its kinetic energy, while the mercury atom with which it collides rises to the first excitation energy level and emits a photon to return to the ground state. The basic condition for such an interaction to occur is that the kinetic energy of the free electrons that will collide with mercury atoms must have a kinetic energy equal to or greater than the first excited energy level of the mercury atom. The free electrons that make such an interaction with mercury atoms either lose all or part of their energy. Therefore, their kinetic energy is either zero or very small compared to other non-colliding electrons. The most important part of the experiment is to distinguish these small kinetic energy electrons from the others and to detect the excitation event. A good solution is to apply a stopping voltage in the opposite direction to the accelerating voltage in order to remove these electrons which have lost

most of their kinetic energy. With the help of this stopping voltage, the free electrons that have lost their kinetic energy, in other words, the free electrons that have succeeded in exciting mercury atoms, can be separated and, accordingly, the excitation regions can be detected.

Free electrons, which have lost most of their kinetic energy, cannot overcome the potential wall created by the applied restraining voltage, but if the accelerating voltage continues to be increased, these electrons stopped at the potential wall may again have sufficient kinetic energy and can overcome this barrier wall.

The most important element of our experimental environment is our furnace with an evacuated discharge tube as shown in Figure 1. The evacuated tube contains a cathode, a perforated anode, a collector electrode, a filament, and an appropriate amount of liquid mercury element at room temperature (25oC). The furnace is heated to a temperature slightly above the vaporization temperature of mercury to provide suitable experimental conditions. The mercury element starts to evaporate at 185°C and the operating temperature is fixed at around 190°C, thus maintaining a constant temperature and keeping the probability of collisions between mercury atoms in the gas phase environment at a constant value, as well as the average free path, which depends on the collisions between free electrons and mercury atoms sent from the cathode into the gas environment.

Another important experimental element in the experimental setup is the rectifier. The uniform electric field, which is required to provide kinetic energy to the free electrons to be used in the process of excitation of mercury atoms and to stop the electrons that can succeed in the excitation process, can only be obtained by applying a uniform voltage. For this reason, this experimental element called a rectifier is used to convert the alternating city voltage into direct voltage. The free electrons that will be used to excite mercury atoms in the experiment are provided by an experimental element called a filament, which is made of a material whose electrons in the outermost orbitals can be easily detached when heated.

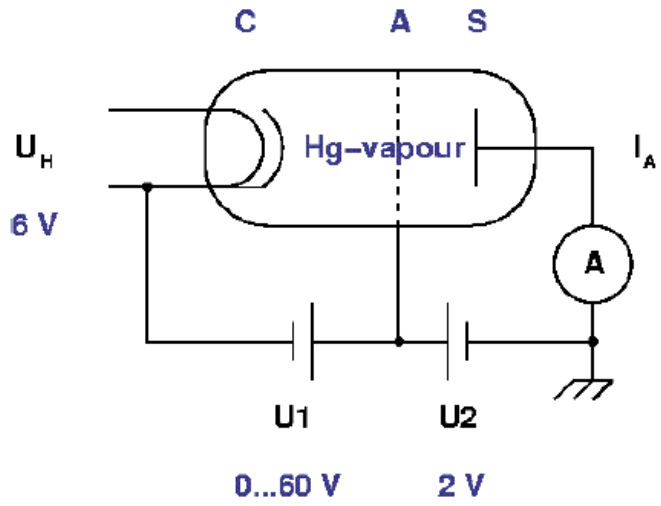


Figure 1: Franck Hertz Tube

With an anode voltage of U1 (4.9 V), the kinetic energy of the electrons is sufficient to bring the valence electron of mercury to the first excitation level 6 $^3P_1$ by inelastic collision. Due to the energy loss, the electron can now no longer travel further in the inverse field between the anode A and the counter electrode S: I current is at a minimum. If the anode voltage is now increased further, the electron kinetic energy is again sufficient to overcome the inverse field: I current intensity increases. When U1 = 4.9 V, two atoms are excited with the same electron because the kinetic energy is high enough: we get a second minimum (Figure 2). This is why the I - U1 graph shows an equal distance between maxima and minima.

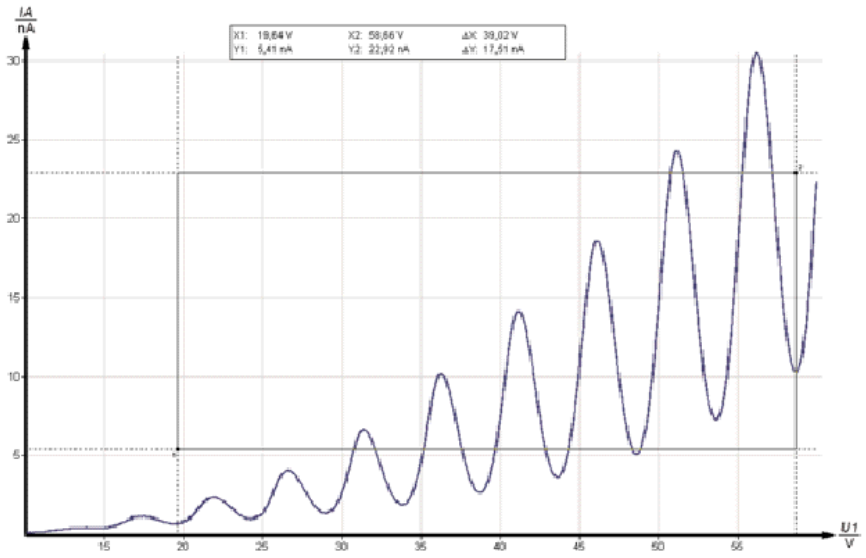


Figure 2: I - U1 graph of the mercury atom.

However, these minima are not well defined because of the initial thermal distribution of electron velocities.

Voltage U1 between anode and cathode $U_1 = U + (\phi_A - \phi_C)$

Here U is the applied voltage, ϕ_A and ϕ_C are the work function voltages of the anode and cathode respectively. Since the excitation energy E is calculated from the voltage differences at minima, the work function voltages are not important here.

According to classical theory, the energy levels that mercury atoms excite can be random. According to quantum theory, however, a defined energy level of the atom appears suddenly.

The trajectory of the I - UA curve was first explained by this and thus a proof of quantum theory was demonstrated.

The excited mercury atom releases the absorbed energy again with the emission of a photon. When the excitation energy E is 4.9 eV, the wavelength of this photon,

$$\lambda = \frac{c\,h}{E} = 253\,nm$$

$$c = 2.9979 \cdot 10^{8}\,\frac{m}{s}$$

$$h = 4.136 \cdot 10^{-15}\,eV$$

This transition is in the UV (ultraviolet) region.

For our calculations, the voltage values of the minima are calculated. By averaging the differences between these values, we find the excitation energy E of the mercury atom.

From the measurements in Figure 2, $E = (4.86 \pm 0.009)eV$

HOW TO CONDUCT THE EXPERIMENT

In this experiment, electrons are accelerated in a tube filled with mercury atoms. The excitation energy of mercury is calculated from the distance between equidistant minima of the electron current in a reversible variable electric field.

A schematic representation of the setup used in the experiment is shown below in Figure 3. The experimental setup consists of a control unit, a Franck-Hertz furnace with an Hg tube, and a computer as shown in Figure 3.



Figure 3: Franck-Hertz experimental setup.

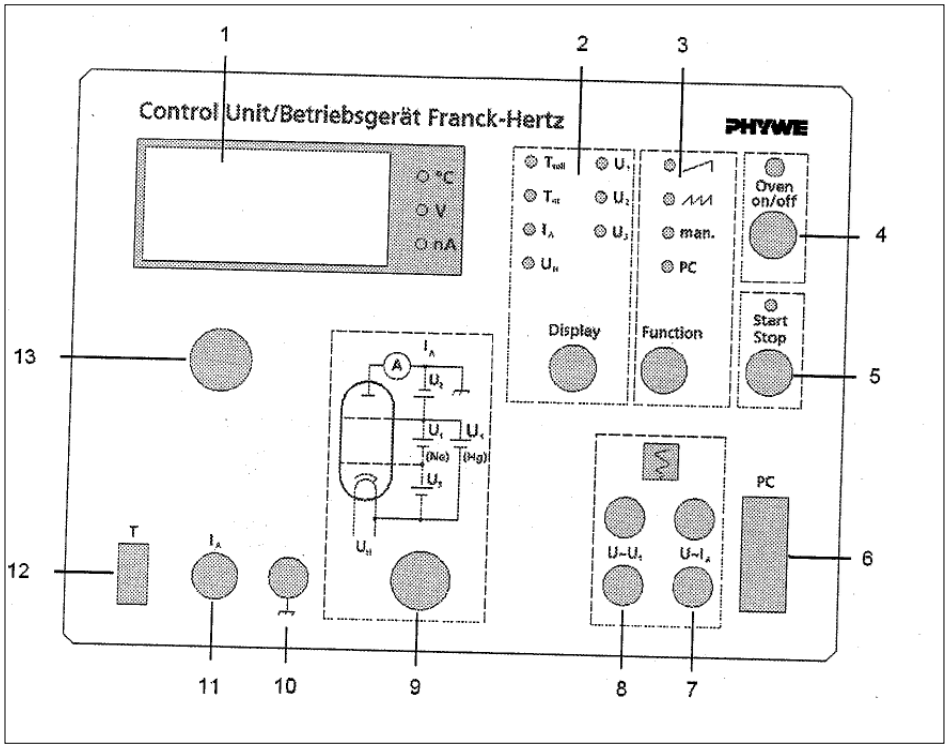


Figure 4: Control unit

Elements of the Control Unit:

1. Three-digit digital display with temperature T, anode current IA, and voltage UH, U1, U2, or U3.

2. "Display" button which allows us to select the attribute to be displayed.

3. "Function" button allowing the selection of "sawtooth", "ramp", "manual control" or "PC control".

4. “Oven on /off” button to activate the heating of the Hg oven.

5. “Start / Stop” button to start or stop the measurement.

6. The socket that serves as the interface for connecting the control unit to the computer.

7. One pair of 4 mm U~IA sockets, Analog output (Y): Voltage proportional to the anode current

8. One pair of 4mm U~IA sockets, Analog output (X): Voltage proportional to the acceleration voltage U1.

9. DIN socket connecting the voltages (UH, U1, U2 and U3) to the tube.

10. GND connector.

11. BNC socket “IA” with input for anode current measurement.

12. Temperature input T. Thermal dissociation socket for DIN plug connection with NiCr-Ni thermal dissociation.

13. Rotary knob for setting the temperature and voltage values.

14. At the rear of the instrument, it provides the voltage for the temperature required for the mercury tube in the Franck-Hertz furnace. Socket with earth connection.

Franck-Hertz Oven:

There is a discharge tube with evacuated air inside. Inside the tube, there is a cathode, a perforated anode, a collector electrode, a filament, and a mercury element at room temperature.

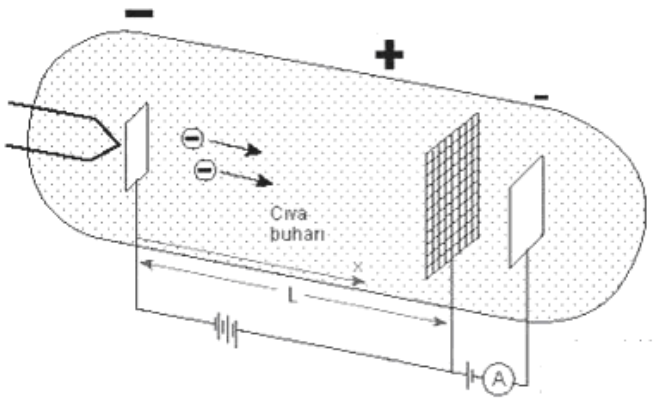


Figure 5: Evacuated Discharge Tube.

Cathode: Produces electrons when heated.

Perforated Anode: Also called grid. It is located opposite the cathode. It is + charged and accelerates the electrons produced.

Collector Electrode: Located behind the Perforated Anode. It collects electrons reaching its surface and creates current. It is slightly negative compared to the perforated anode to ensure that only electrons that can exceed a certain energy level can reach it.

Flame: Made of a material whose electrons in the outermost orbitals can be easily detached when heated, it is the experimental element that generates the free electrons used to excite the mercury atom.

At a certain voltage (U1 = U2), which depends on temperature, a reddish charge-discharge occurs between the anode and cathode by ionization. Therefore, meaningful measurements can only be taken at voltages U1 < U2.

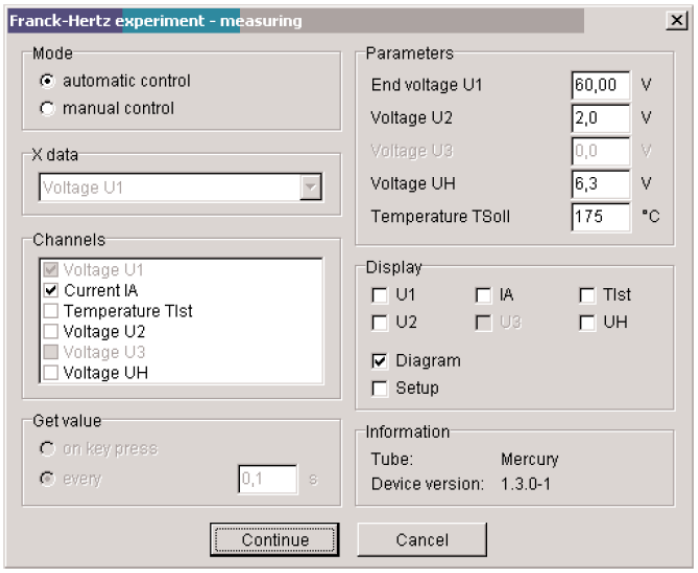


Figure 6: Measurement parameters.

1. Set up the experimental set as shown in Figure 3. Connect the operating unit to the computer.

2. Turn on the control unit and set it to Cobra PC.

3. Start the measurement program and select the Cobra3 Franck-Hertz tester. Enter the parameters given in Figure 6 into the computer and press continue. Now the oven of the Franck-Hertz tube will be heated to 175 ^{0}C and the graph between U1 and IA will be observed as shown in Figure 2.

4. Then enter the U1 values into the computer. Read the IA values and record them in Table 1, draw the graph between U1 and IA by hand, and find the 1st excitation energy of the mercury atom.

5. Find the wavelength of the emitted photon using the excitation energy and determine in which region of the electromagnetic spectrum it lies.

Figure 2: Franck-Hertz curve recorded at T=175 ^{0}C and U2 = 2V.

Table 1: Results

MAX		**MIN**	
U_1 (v)	I_A(nA)	U_1 (v)	I_A(nA)

REFERENCES

1. H Haken, H C Wolf, The Physics of Atoms and Quanta, 2000, Springer
2. Atoms, Molecules and Photons, Demtröder, Springer 2005
3. Phywe experimental literature and operating manuals
4. http://spiff.rit.edu/classes/phys314/lectures/fh/fh.html
5. http://hyperphysics.phy-astr.gsu.edu/hbase/FrHz.html